Observer-Patch Holography as a String-Vacuum Selector: Observers, Clocks, Edge Strings, and the Bouchard-Donagi Candidate

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Abstract

String theory is read here as an effective language for OPH edge dynamics. Finite observer patches compare shared records, and their settled repair histories can be read as worldsheet-like histories on the edge system.

The paper uses that reading as a vacuum-selection sieve. OPH supplies the target spacetime, Standard Model structure, records, interfaces, checkpoints, and synchronization rules. A conventional string vacuum is promoted only after the geometric, worldsheet-critical, threshold, and moduli-locking gates pass. The worked object is a heterotic Bouchard–Donagi type Standard Model candidate with one massless Higgs pair and explicit safety gates. The published construction supplies a visible massless-cohomology spectrum, without the stabilized nonzero-mode spectrum or decoupling map needed for a low-energy threshold calculation. The threshold/spectrum certificate is not emitted here, so this paper does not promote the candidate to a passing OPH-correct witness.

What This Paper Contributes

String theory supplies a large language of edge dynamics, compactification, moduli, and Standard Model candidates. This paper reads that language from the OPH side. The OPH core first fixes the target spacetime, gauge quotient, one-Higgs branch, records, clocks, and observer-facing boundary data. The string description then becomes an effective edge language for histories constrained by OPH.

The sieve carries the interaction package with the particle spectrum: the OPH Einstein branch supplies the gravity target, the compact-gauge branch supplies the strong/electroweak target, and the exact Standard Model quotient fixes the visible charge lattice. The string representative must realize those forces and the particle content on the same observer-visible branch.

The worked contribution is a sieve for string-vacuum candidates. A Bouchard–Donagi type heterotic object is treated as a named candidate and tested against OPH gates: one massless Higgs pair, the Standard Model quotient, operator safety, worldsheet criticality, threshold compatibility, and moduli locking. Every unemitted certificate is explicit. A vacuum survives only if its string data land on the OPH quantitative vector without retuning. The statement selects string data through OPH gates; it does not promote a reference ensemble or ordinary string vacuum into the OPH-native vacuum.

Contents

1 Scope and Claim Boundary

This paper records a specific bridge from the de Sitter observer problem to the OPH string continuation. It separates three layers:

$$
\begin{aligned}
&\text{observer/clock implementation} \\
&\quad \longrightarrow \text{edge-string effective language} \\
&\quad \longrightarrow \text{critical string representative.}
\end{aligned}
$$

The first layer is OPH fixed-cutoff patch algebra. The second layer is the heat-kernel edge-sector read-off. The third layer is the Bouchard-Donagi heterotic Standard Model used as a named heterotic zero-mode candidate. The heat-kernel edge identity by itself does not prove a critical worldsheet CFT; the critical-edge certificate below is an additional condition on the continuum edge branch.

1.1 Forces, gauge group, and effective strings

The reader-facing phrase "particles and forces" is translated here into OPH gates. The particle package is not independent of the interaction package: matter representations, charges, gauge bosons, and the graviton are all read from the same observer-visible normal form.

Gravity. The OPH target is a single Einstein-frame metric perturbation on the physical quotient, with one massless transverse-traceless graviton. The critical representative must identify its closed-string symmetric spin-two state with that OPH graviton and match $G_4^{\text{string}} = G_4^{\text{OPH}}$.

Strong force. The color factor is the $\mathrm{SU}(3)_c$ part of the global quotient $(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6$. The visible compactification must realize this charge lattice with no light chiral exotics.

Weak and electromagnetic forces. The electroweak target is the $\mathrm{SU}(2) \times \mathrm{U}(1)$ branch, the exact hypercharge lattice, the one-Higgs target, and the resulting low-energy $U(1)_{\text{em}}$. The candidate must land on that one-Higgs Standard Model branch and preserve the OPH hypercharge quotient. The local Borel–Weil model $H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$ is used only for the observed light one-Higgs electroweak projection; it does not replace the full supersymmetric Higgs-pair bookkeeping in a string witness.

Standard Model gauge group. The target is the global group

$$G_{\text{phys}} = (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6,$$

including the finite quotient. The Bouchard-Donagi Wilson-line centralizer must equal $S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong G_{\text{phys}}$, excluding simple-GUT X/Y gauge bosons from the unbroken visible adjoint.

Effective string theory. The OPH input is stable cyclic edge normal forms, collar sewing, records, and accepted repair histories. The fixed-cutoff read-off is $Z_{\text{edge}} = K_t(1)$; a controlled large-edge branch gives the perturbative worldsheet language, and the critical-edge certificate decides whether the branch is a heterotic critical string.

Thus the paper does not add forces after selecting particles. It first fixes the observer-visible interaction structure and then asks which string compactification realizes that whole structure.

The three Standard Model gauge forces are the low-energy decomposition of the global quotient, while gravity is the OPH Einstein branch read as the closed-string graviton only after a critical lift.

The target claim is conditional. If the OPH recovered-core theorem stack is accepted, if the Bouchard-Donagi one-Higgs stratum locks to the OPH quantitative vector without free retuning, and if the MSSM $\mathbb{Z}_4^R$ safety layer is realized on the same branch, OPH becomes an independent selector of a particular known heterotic Standard Model representative:

$$BD^{\mathrm{OPH}}_{n=1,+}.$$

The notation separates two roles:

$$BD^{\mathrm{OPH}}_{n=1} \quad \text{is the geometric one-Higgs witness,}$$

$$BD^{\mathrm{OPH}}_{n=1,+} = BD^{\mathrm{OPH}}_{n=1} + \mathbb{Z}_4^R \quad \text{is the operator-safe candidate.}$$

The $+$ records the MSSM $\mathbb{Z}_4^R$ safety layer. The charge-algebra part of this gate closes inside the paper; the realization of $\mathbb{Z}_4^R$ as a symmetry of the BD compactification is a certificate gate.

The compact rule is:

$$\text{string theory is the edge-language of OPH.}$$

In that reading, the string landscape is the space of effective edge-worldsheet completions. OPH acts as a selector by imposing the observer-visible normal form.

Goal	**Paper claim**	**Receipt carried here**
Named string candidate	OPH tests the operator-safe Bouchard–Donagi one-Higgs heterotic Standard Model candidate $BD^{\mathrm{OPH}}_{n=1,+}$. Passing OPH-correct status requires the unemitted threshold/spectrum and other certificate gates.	Selector gates, global group proof, safety charge table, candidate sieve.
String-theory pressure points	The landscape, global group, Higgs multiplicity, exotics, operator danger, proton decay, and moduli problems become explicit gates.	Gate table, acceptance table, falsifier matrix.
Empirical test surface	The paper defines low-energy and threshold proxy targets against which a completed branch could fail.	Higgs/top target coordinates, stop proxy, one-loop gauge-running proxy, operator table.

2 OPH Entry Map for String Theorists

For a reader entering OPH from string theory, the shortest safe translation is this. OPH begins with finite observer patches. Each patch has local data, records, and overlap readouts. A physical world is the quotient-normal form reached when accepted repair moves lower mismatch and the local-diamond plus repair-completeness conditions make the observer-facing result schedule-independent from a fixed initial quotient state. Same-boundary uniqueness requires a preserved boundary or sector with a unique consistent quotient extension. Geometry, gauge structure, particles, and edge strings are read from that normal form.

2.1 Five informal summaries

1. The basic rule. Observer patches are finite. They see shared boundary data. Physics is the public content that survives overlap comparison and repair. The synthesis paper gives the broad entry point [1]; the consensus paper gives the fixed-cutoff repair theorem surface [3].

2. Why spacetime and gauge theory appear. Null structure, Lorentzian geometry, and the Einstein branch are recovered on the stated observer-overlap consistency surface. In the gauge lane, overlap/holonomy data classify zero-obstruction transportable sectors, DR/Tannaka reconstruction gives a compact group from that category, and Minimal Admissible Realization (MAR) plus the explicit matter package selects the realized Standard Model quotient. Thus obstruction cancellation supplies transportability and classification. Standard Model selection requires MAR and the explicit matter package. The compact paper carries the theorem surface for this claim [2]. The Yang-Mills note states the compact-gauge branch in the language of holonomy, Euclidean transfer, and the mass-gap problem [9].

3. Why particle numbers enter the selector. The particle-sector paper carries the electroweak, Higgs/top, quark, charged-lepton, neutrino, and hadron audit lanes [5]. The fine-structure note isolates the local pixel fixed point that feeds the quantitative electroweak target [6]. These papers supply the numerical vector against which the named candidate branch is screened.

4. Why observers and records are literal inputs. The screen-microphysics paper turns patches into finite record-bearing carriers with ports, interfaces, synchronization rules, and checkpoint readouts [4]. The thinking paper shows the same patch-net fixed-point machine as a model of biological cognition [10]. For this string paper, the needed fact is concrete: observers are record-bearing finite systems inside the theory.

5. Why phenomenology becomes a gate list. The dark-matter paper treats missing gravitational response as a quotient-edge scalar-channel phenomenon with explicit stress-parent gates [7]. The χ_ν note records the coherent-matter source generator and susceptibility bounds used by that collar branch [8]. The metaphysics continuation layer is background for the fixed-point ontology and is outside the string selector proof.

2.2 Source map

The OPH reader path used here has ten entries. Each link points to the TeX source in the public GitHub repository.

Source	Role in this paper	GitHub
Observers Are All You Need	Synthesis entry point: finite observer cuts, overlap agreement, screen-capacity language, and the recovered-branch map.	paper/source

Recovering Relativity and the Standard Model from Observer Overlap Consistency	Compact theorem surface for Lorentz, Einstein, zero-obstruction compact-gauge reconstruction, MAR-selected global Standard Model quotient, hypercharge, colors, generations, and no massless mixed X/Y generator in the connected visible adjoint.	paper/source
Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics	Fixed-cutoff consensus theorem: mismatch functional, accepted repair, quotient confluence, and holonomy obstructions.	paper/source
Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH	Finite carrier model for records, ports, interfaces, checkpoint restoration, and observer synchronization.	paper/source
Deriving the Particle Zoo from Observer Consistency	Particle and electroweak quantitative lanes used by the moduli-locking target vector.	paper/source
The Fine-Structure Constant as an OPH Pixel Fixed Point	Local screen-cell fixed point and fine-structure input used by the electroweak target.	extra/source
Observer-Patch Holography and the Dark Matter Phenomenon	Quotient-edge scalar-channel branch, used as a phenomenology example of OPH gate logic.	cosmology/source
Theoretical Bounds on χ_ν in Observer-Patch Holography	Coherent-matter source generator, collar survival, and susceptibility bounds for the dark-sector continuation.	extra/source
Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics	Compact-gauge reconstruction and mass-gap route, useful for the gauge side of the string selector.	extra/source
Thinking as Patch-Net Fixed-Point Search	Cognitive instance of the same finite patch-net record and repair machine.	extra/source

3 Main Result: Conditional BD Witness Selection up to OPH Equivalence

An OPH-correct critical string is defined only after the equivalence relation has been fixed. The claim concerns observer-visible physical data, not notation, duality frame, worldsheet gauge, or coordinate presentation. This paper supplies a named BD witness target and a finite encoded audit set. A global singleton statement requires an exhaustive candidate-class certificate in addition to the witness certificates.

Definition 3.1 (OPH-equivalent critical presentations)**.** *Two critical-string presentations $\mathcal{T}_1$ and $\mathcal{T}_2$ are OPH-equivalent, written*

$$\mathcal{T}_1 \sim_{\mathrm{OPH}} \mathcal{T}_2,$$

when they induce the same observer-visible data:

1. *the same edge-sewn heat-kernel partition on the compact gauge branch;*

2. *the same four-dimensional graviton normalization and Einstein-frame metric perturbation;*

3. *the same global visible gauge group*

$$G_{\mathrm{phys}} = (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6;$$

4. *the same hypercharge lattice, chiral index, and low-energy matter package;*

5. *the same operator-safety algebra on the visible superpotential;*

6. *the same OPH quantitative target vector $\mathcal{O}_{\mathrm{OPH}}$, modulo threshold and moduli coordinates invisible to the OPH quotient.*

Duality frames are charts on the same OPH-visible normal form.

Definition 3.2 (OPH-correct critical string)**.** *A critical-string presentation $\mathcal{T}$ is OPH-correct if it satisfies all of the following.*

1. *a critical completion of the OPH sewn-edge worldsheet branch, with the same observer-visible edge data.*

2. *Its massless spin-two field is the quantization of the OPH Einstein-frame metric mode.*

3. *Its visible compact gauge sector descends to $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.*

4. *Its visible chiral spectrum is exactly three generations, no light chiral exotics, and the one-Higgs-pair branch used by the OPH electroweak target.*

5. *Its visible operator algebra has the $\mathbb{Z}_4^R$ safety layer, or an OPH-equivalent rule, which permits Yukawas and the Weinberg operator, and forbids perturbative RPV, perturbative dimension-five proton decay, and the perturbative μ-term.*

6. *Its moduli and threshold map satisfies*

$$\mathcal{F}_{\mathcal{T}}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$$

with full transverse rank after quotienting OPH-invisible redundancies.

Theorem 3.3 (Conditional BD witness theorem)**.** *Assume the OPH recovered-core target, the heterotic edge-sector branch including the finite-carrier critical-edge certificate, and the BD certificate gates listed in Section 18. Assume also that the Bouchard–Donagi one-Higgs branch with the operator-safety layer satisfies the cohomology, global-group, safety, threshold, and moduli-locking gates, including a certified low-energy decoupling map. Then*

$$\boxed{BD^{\mathrm{OPH}}_{n=1,+}}$$

is a passing named OPH-correct critical-string witness, modulo $\sim_{\mathrm{OPH}}$.

Proof. The recovered-core target, after MAR selection of the admissible one-Higgs branch, fixes the observer-visible endpoint: the global Standard Model quotient, three colors, three generations, exact hypercharge lattice, one-Higgs electroweak branch, no light chiral exotics, no extra visible low-scale U(1), and the product-group absence of massless X/Y gauge bosons. The edge-string continuation restricts the witness to critical presentations whose worldsheet branch is the sewn

OPH edge system and whose massless spin-two mode is the quantized OPH metric perturbation. The Bouchard–Donagi $\mathbb{Z}_2$-quotient SU(5)-bundle with one Higgs pair supplies the named heterotic witness with the required three-generation, no-exotics visible massless cohomology and Wilson-line centralizer. The $\mathbb{Z}_4^R$ layer closes the visible operator gate. The assumed threshold/spectrum and decoupling certificate supplies the physical low-energy map. The moduli-locking hypothesis isolates the physical target point for that witness. Those conditions are exactly the OPH-correctness gates for the named witness. Quotienting by $\sim_{\text{OPH}}$ removes pure presentation duplicates of the same observer-visible data. □

Proposition 3.4 (Global singleton certificate criterion)**.** *Let $\mathcal{C}$ be an explicitly enumerated candidate class containing every critical-string presentation allowed to compete for the same OPH-visible target. Suppose every row of $\mathcal{C}$ has a reproducible gate certificate, and suppose the certified survivors are ordered first by gate satisfaction, then by the minimality vector*

$$C(v) = (N_{\text{parent}}, N_{\text{corridor}}, |\Gamma|, n_H, N_{\text{exotic}}, N_{U(1),\text{extra}}, N_{\text{safety}}, D_{\text{moduli}}),$$

with ties quotienting by $\sim_{\text{OPH}}$. If the only row in $\mathcal{C}/\sim_{\text{OPH}}$ that passes all gates and attains the minimality bound is $BD^{\text{OPH}}_{n=1,+}$, then the singleton statement holds for that enumerated class:

$$\boxed{\{\mathcal{T} \in \mathcal{C} : \mathcal{T} \text{ is OPH-correct}\}/\sim_{\text{OPH}} = \{BD^{\text{OPH}}_{n=1,+}\}.}$$

Proof. This is the finite-selector normal-form argument. The enumerated certificate table supplies the candidate set, the gate vector, and the minimality vector for every row. The hypothesis says that all rows outside $BD^{\text{OPH}}_{n=1,+}$ either fail a gate or lose the minimality comparison after quotienting by $\sim_{\text{OPH}}$. Therefore the certified survivor set in $\mathcal{C}/\sim_{\text{OPH}}$ is the stated singleton. Without the exhaustive table for $\mathcal{C}$, this proposition is a certificate criterion rather than a global landscape theorem. □

Remark 3.5 (How strong this is)*.* BD is not decoration here. OPH supplies an external observer-visible target, so a named branch can be tested by a normal-form equation. The claim is falsifiable by construction: failure of the BD cohomology, safety-layer realization, threshold/spectrum certificate, decoupling map, or transverse moduli rank retracts the named witness without retracting the OPH recovered core. Global uniqueness is a separate comparative-catalogue certificate.

4 Theorem Agenda

The de Sitter observer literature has isolated a tight set of frontier puzzles. OPH's job is to make them concrete. The de Sitter side of the agenda is:

Pressure point	OPH theorem target
Physical observer in de Sitter	Finite observer theorem: a physical observer is a stable record-bearing patch subfederation.
Complex semiclassical correlators	Clock-projection correlator theorem: imaginary phases appear after record-sector clock projection.
Time-reversal holonomy	$\mathbb{Z}_2^T$ holonomy theorem: clock flips are cycle obstructions, equivalently $w_1(L_T)$ classes.
Time-reversal breaking	Record-branch theorem: ordered checkpoint records select a local time orientation.
Entropy location	Edge-capacity theorem: $S_{\mathrm{dS}} = A/(4\ell_P^2)$ is finite edge-center record capacity.
Finite de Sitter degrees	Static-patch matrix-carrier theorem: finite patch and interface algebras supply pre-geometric matrix degrees.
DSSYK/QCD/string parallels	Edge-sum universality theorem: heat-kernel edge sums reorganize as large-N worldsheet expansions.
Critical worldsheet closure	Heterotic edge-polarization gate: the sewn-edge scaling limit must realize $(\mathrm{Heis}_8 \otimes V_{\Gamma_{16}})_L \otimes (\mathrm{Heis}_8 \otimes \mathrm{Ferm}_8)_R$, with no residual coset.

The string/vacuum side of the agenda is:

Pressure point	OPH theorem target
String landscape	OPH vacuum sieve theorem: critical completions must hit the observer-visible normal form.
Worldsheet criticality	OPH edge-polarization theorem: the sewn-edge scaling limit must be identified with the heterotic edge VOA, including OPEs, grading, spin structures, and no residual coset.
Global Standard Model group	Global group locking theorem: $S(\mathrm{U}(3)\times\mathrm{U}(2)) \cong (\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6$.
Massless unified-gauge carriers	Connected-adjoint theorem: the massless product-group adjoint contains no mixed $(3, 2, \pm 5/6)$ generator; massive carriers remain a spectrum gate.
Moduli stabilization	OPH moduli-locking criterion: $\mathcal{F}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$ with full transverse rank.

5 De Sitter Observer Pressure Points

Two de Sitter papers make the observer problem unusually explicit. One note relates the need for observers in de Sitter space to spontaneous breaking of time reversal [11]. A follow-up states that quantum de Sitter theory is widely thought to require a physical observer in the static patch, with the definition of observer unspecified [12]. The time-reversal question is sharpened by treating time reversal as a gauge symmetry hidden by spontaneous symmetry breaking, with a proposed "smoking gun" given by a closed curve whose holonomy flips forward-going clocks into backward clocks [13].

OPH reads these as implementation questions. What finite object counts as an observer? What operation selects the clock branch? What finite obstruction carries the clock flip? What entropy surface supplies the static-patch degrees of freedom?

5.1 Observer

Definition 5.1 (OPH observer subfederation)**.** *Let*

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}, \{\mathcal{I}_e\}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\})$$

be a finite OPH patch carrier. An observer subfederation is a finite $U \subset V$ equipped with an accessible algebra $\mathcal{A}_U$, a state ρ_U, a record algebra $\mathcal{R}_U$, a boundary interface $\mathcal{I}_{\partial U}$, and accepted repair maps whose normal form preserves a checkpoint order on $\mathcal{R}_U$.

The observer projection is the record event

$$P_{\mathrm{obs}} = P_{\mathcal{R}_U,+},$$

where $+$ denotes the chosen clock orientation. Conditioning is ordinary projection:

$$\rho|_{\mathcal{R}_U,+} = \frac{P_{\mathcal{R}_U,+}\rho P_{\mathcal{R}_U,+}}{\mathrm{Tr}(P_{\mathcal{R}_U,+}\rho)}.$$

Theorem 5.2 (Finite observer theorem)**.** *On a finite OPH static-patch carrier, a physical observer is a subfederation $U \subset V$ whose checkpoint*

$$\mathrm{Chk}_U(t) = \big(\mathcal{R}_U(t), \rho_U^{\mathrm{acc}}(t), \mathfrak{I}_U^{\mathrm{ext}}(t), \nu_{\geq t}, \mathfrak{B}_U(t)\big)$$

has a stable continuation law on the observer-accessible event algebra under same-interface continuation. The de Sitter observer projection is the record-sector projection $P_{\mathcal{R}_U,+}$.

Proof. The data defining U are finite algebraic data. The event that records persist with a monotone checkpoint ordering is a projection in the record algebra after passing to the observable quotient. Conditioning by that projection gives the usual post-selected state on the observer-accessible sector. The observer is represented inside the patch Hilbert space. □

5.2 Semiclassical De Sitter

OPH treats semiclassical spacetime as an observer-facing quotient normal form. The repair functional has the schematic form

$$\Phi(s) = \sum_{e=\{i,j\}} w_e\, d_e(\pi_{i,e}(s_i), \pi_{j,e}(s_j))\,.$$

Accepted repairs lower Φ. Under the local-diamond and repair-completeness hypotheses used in the consensus paper, the terminal observable state is schedule-independent on the physical quotient.

Claim 5.3 (Semiclassical patch)**.** *The semiclassical de Sitter patch is the stable quotient normal form of finite observer-overlap repair, read on a record-bearing clock sector.*

OPH answers the de Sitter observer question here. The maximally mixed static-patch state belongs to the unconditioned algebra. The semiclassical branch is read after record conditioning.

5.3 Clock-conditioned correlator

The finite toy model used in the correspondence package has two clock orientations exchanged by time reversal. The two-level model is only a witness. The algebraic point is that the maximally mixed static-patch trace averages over clock orientations, and an observer record projection selects one oriented branch.

Theorem 5.4 (Clock-projection correlator theorem)**.** *Let $\mathcal{H}$ be finite-dimensional, let $H = H^\dagger$, let $A = A^\dagger$, and set*

$$A(t) = e^{iHt} A e^{-iHt}, \qquad \rho_\infty = I/d.$$

Then

$$C_\infty(t) := \mathrm{Tr}(\rho_\infty A(t) A)$$

is real, so

$$\mathrm{Im}\, C_\infty(t) = 0.$$

Equivalently,

$$\mathrm{Tr}(\rho_\infty [A(t), A]) = 0.$$

For a noncentral observer-clock record projector P_E,

$$\rho_E = \frac{P_E \rho_\infty P_E}{\mathrm{Tr}(P_E \rho_\infty)},$$

the commutator expectation is generically nonzero. In the two-state clock example

$$H = \frac{\omega}{2}\sigma_z, \qquad A = \sigma_x,$$

one has

$$C_\infty(t) = \frac{1}{2}\,\mathrm{Tr}(\sigma_x(t)\sigma_x) = \cos(\omega t).$$

On the forward clock record state $\rho_+ = |0\rangle\langle 0|$,

$$C_+(t) = \langle 0|\sigma_x(t)\sigma_x|0\rangle = e^{i\omega t}.$$

The imaginary semiclassical phase is restored by clock-record conditioning:

$$\mathrm{Im}\, C_\infty(t) = 0, \qquad \mathrm{Im}\, C_+(t) = \sin(\omega t).$$

Proof. Since $A(t)$ and A are Hermitian,

$$C_\infty(t)^* = \frac{1}{d}\,\mathrm{Tr}(AA(t)).$$

By cyclicity of the trace,

$$\mathrm{Tr}(AA(t)) = \mathrm{Tr}(A(t)A),$$

so $C_\infty(t)^* = C_\infty(t)$, hence $C_\infty(t) \in \mathbb{R}$. The commutator statement follows from $\mathrm{Tr}([A(t), A]) = 0$.

For a projected state ρ_E, the projector sits between the state and the clock-transition operator. The cyclic cancellation above applies only in the commuting case. Generic observer-clock records are noncentral, so the antisymmetric part can survive. For the two-state witness,

$$\sigma_x(t) = e^{i\omega t\sigma_z/2}\sigma_x e^{-i\omega t\sigma_z/2} = \cos(\omega t)\sigma_x - \sin(\omega t)\sigma_y.$$

Then

$$\sigma_x(t)\sigma_x = \cos(\omega t) I + i\sin(\omega t)\sigma_z.$$

The maximally mixed trace gives $\cos(\omega t)$. The $|0\rangle$ matrix element gives $\cos(\omega t) + i\sin(\omega t) = e^{i\omega t}$. □

5.4 Time-reversal holonomy

The time-orientation data must be a local system. Globally chosen patch signs give only a coboundary, and every closed-cycle product then telescopes to $+1$. Clock-flip holonomy needs a $\mathbb{Z}_2^T$-valued Čech 1-cocycle on the overlap nerve $N(\mathfrak{F})$:

$$g_{ij} \in \mathbb{Z}_2^T, \qquad g_{ij}g_{jk}g_{ki} = 1$$

on triple overlaps, modulo local redefinitions

$$g_{ij} \sim \lambda_i g_{ij} \lambda_j^{-1}, \qquad \lambda_i \in \mathbb{Z}_2^T.$$

For a closed path $\gamma = (i_0 i_1)(i_1 i_2)\cdots(i_{n-1} i_0)$, define

$$h_T(\gamma) = \prod_{a=0}^{n-1} g_{i_a i_{a+1}} \in \mathbb{Z}_2^T.$$

Theorem 5.5 (Time-reversal holonomy theorem)**.** *A forward-going local clock transported around γ returns backward-going iff*

$$h_T(\gamma) = -1.$$

The obstruction to a global time-orientation section is the cohomology class

$$[g] \in H^1(N(\mathfrak{F}), \mathbb{Z}_2) \cong \check{H}^1(N(\mathfrak{F}), \mathbb{Z}_2),$$

equivalently the first Stiefel-Whitney class

$$w_1(L_T) = [g].$$

It vanishes iff the transition cocycle is a coboundary, i.e. iff one can choose local orientations ϵ_i with $g_{ij} = \epsilon_i^{-1}\epsilon_j$ on every overlap.

Proof. Parallel transport across the overlap (ij) multiplies the clock-orientation basis by g_{ij}. Transport around γ multiplies by the ordered product $h_T(\gamma)$. Since the group is $\mathbb{Z}_2$, the value $+1$ preserves orientation and the value -1 reverses it.

A global time orientation is a choice of signs $\epsilon_i \in \mathbb{Z}_2$ such that the transition from patch i to patch j is reproduced by $g_{ij} = \epsilon_i^{-1}\epsilon_j$. This is the statement that the 1-cocycle g is a coboundary. The obstruction class is $[g] \in H^1(N(\mathfrak{F}), \mathbb{Z}_2)$, which is the first Stiefel-Whitney class of the associated $\mathbb{Z}_2$ time-orientation line bundle L_T. A nonzero class means no global time-orientation section exists. □

The proposed de Sitter clock-flip test reads, in finite OPH form, as a $\mathbb{Z}_2^T$ cycle obstruction on the patch-overlap nerve.

5.5 Record-branch time-reversal breaking

Let $P_1, P_2, \ldots, P_n \in \mathcal{R}_U$ be mutually commuting checkpoint record projectors over a finite observation window, with a monotone checkpoint order

$$P_1 \prec P_2 \prec \cdots \prec P_n.$$

Time reversal sends this ordered history to the reversed history. Conditioning on the finite branch projector

$$P_{\text{branch}} = P_1 P_2 \cdots P_n$$

selects one local time orientation on the observer-accessible quotient.

Theorem 5.6 (Record-branch time-reversal breaking)**.** *The unconditioned finite carrier carries the time-reversal-related pair of record orders. A stable checkpoint branch selects one order by*

$$\rho \mapsto \frac{P_{\mathrm{branch}}\rho P_{\mathrm{branch}}}{\mathrm{Tr}(P_{\mathrm{branch}}\rho)}.$$

Time reversal is a redundancy of the unconditioned carrier and is hidden on the conditioned observer quotient by finite record ordering.

Proof. Because the checkpoint records belong to the observer-accessible record algebra, they are central or mutually commuting on the exact fixed-cutoff event surface used here. The finite product $P_{\mathrm{branch}} = P_1 \cdots P_n$ is again a projector onto the event that the ordered record history occurs. The time-reversed branch is represented by the same unordered product together with the opposite continuation schedule. The unconditioned carrier contains both orientation-related continuations. Conditioning by P_{branch} and the continuation schedule selects one law on the observer-accessible quotient. On that quotient the reversed branch is outside the conditioned continuation law, so the time-reversal redundancy is hidden by record ordering. □

6 Entropy, Matrix Degrees, and Scale Separation

DSSYK/JT-de Sitter work places the stretched-horizon entropy at order Planck distance from the mathematical horizon [14]. OPH agrees with that direction. The entropy budget is finite edge-center and record capacity:

$$S_{\mathrm{dS}} = N_{\mathrm{scr}} = \frac{A_{\mathrm{dS}}}{4\ell_P^2}.$$

The string scale is downstream in this paper. The edge record capacity is the substrate; the worldsheet is the effective language after sewing and large-edge organization.

Theorem 6.1 (Edge-capacity entropy theorem)**.** *On the OPH de Sitter static-patch branch, the horizon entropy is the capacity of the observer-facing edge-center record surface:*

$$S_{\mathrm{dS}} = N_{\mathrm{scr}} = \frac{A_{\mathrm{dS}}}{4\ell_P^2}.$$

The corresponding screen-capacity branch gives

$$\Lambda = \frac{3\pi}{GN_{\mathrm{scr}}}.$$

Proof. The OPH screen-capacity branch identifies the number of observer-facing horizon record units with the Gibbons-Hawking entropy,

$$N_{\mathrm{scr}} = \frac{A_{\mathrm{dS}}}{4\ell_P^2}.$$

For a four-dimensional de Sitter static patch,

$$A_{\mathrm{dS}} = 4\pi R_{\mathrm{dS}}^2, \qquad \Lambda = \frac{3}{R_{\mathrm{dS}}^2}.$$

This gives

$$N_{\mathrm{scr}} = \frac{4\pi R_{\mathrm{dS}}^2}{4\ell_P^2} = \frac{\pi R_{\mathrm{dS}}^2}{\ell_P^2}.$$

Using $\ell_P^2 = G$ in units $\hbar = c = 1$,

$$N_{\rm scr} = \frac{3\pi}{G\Lambda},$$

and

$$\Lambda = \frac{3\pi}{GN_{\rm scr}}.$$

□

The finite carrier

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}, \{\mathcal{I}_e\}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\})$$

also supplies a matrix-style pre-geometric object. This matches the pressure behind de Sitter-matrix arguments, where black holes and nonperturbative sectors probe the static-patch degrees of freedom [15]. In OPH, black-hole sectors are constrained normal forms that reallocate screen capacity:

$$P(\text{sector}) \propto \exp[-(S_{\rm dS} - S_{\rm sector})].$$

Theorem 6.2 (Static-patch matrix carrier theorem). *The finite algebras $\mathcal{A}_i$ and interface algebras $\mathcal{I}_e$ of an OPH de Sitter static-patch carrier are the microscopic matrix degrees used by the pre-geometric description. A black-hole sector is a constrained normal-form sector $\mathfrak{S}_{\rm BH}$ with*

$$S(\mathfrak{S}_{\rm BH}) < S_{\rm dS}$$

and fluctuation weight proportional to

$$\exp[-(S_{\rm dS} - S(\mathfrak{S}_{\rm BH}))].$$

Proof. At fixed cutoff each local patch algebra and interface algebra is finite-dimensional, hence a finite direct sum of matrix algebras. The unreduced tensor product of a finite static-patch carrier is a finite matrix algebra up to superselection-block decomposition, and the physical algebra is its overlap-invariant quotient. A constrained black-hole sector imposes additional horizon, energy, or area constraints on the same finite record surface, so its accessible record count is smaller than the empty de Sitter static-patch count. With the same coarse-grained entropy measure in both macroscopic sectors, the relative weight of the constrained sector is the ratio of state counts,

$$\frac{e^{S(\mathfrak{S}_{\rm BH})}}{e^{S_{\rm dS}}} = \exp[-(S_{\rm dS} - S(\mathfrak{S}_{\rm BH}))].$$

□

Scale-separation work on de Sitter and double-scaled SYK distinguishes a cosmic sector from a microscopic sector [16]. OPH has the same split:

$$\text{cosmic sector} \leftrightarrow N_{\rm scr}, \qquad \text{microscopic sector} \leftrightarrow P,\ \text{edge labels, finite records.}$$

The clean dimensionless separator is $N_{\rm scr}/P$.

7 Edge-String Emergence

The OPH string continuation begins at the edge-sector partition. At fixed cutoff, edge labels are compact representations R with dimension d_R and quadratic Casimir $C_2(R)$. The open-edge weight is

$$p_R(t) = \frac{d_R e^{-tC_2(R)}}{Z_{\text{open}}(t)}.$$

Sewing two collar sides contributes a second representation-dimension factor:

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}.$$

By Peter-Weyl,

$$K_t(g) = \sum_R d_R \chi_R(g) e^{-tC_2(R)}.$$

At $g = 1$, $\chi_R(1) = d_R$, hence

$$Z_{\text{edge}}(t) = K_t(1).$$

Theorem 7.1 (OPH edge-string read-off)**.** *The sewn OPH collar partition equals the compact-group heat kernel at the identity:*

$$Z_{\text{edge}}(t) = K_t(1).$$

On a large-N_{edge} branch with $g_s = 1/N_{\text{edge}}$, the controlled surface expansion has the closed-string genus form.

Proof. The compact heat kernel has Peter-Weyl expansion

$$K_t(g) = \sum_R d_R \chi_R(g) e^{-tC_2(R)}.$$

At the identity, $\chi_R(1) = d_R$, so

$$K_t(1) = \sum_R d_R^2 e^{-tC_2(R)}.$$

A closed OPH collar is obtained by sewing two open edge boundaries, and the second boundary contributes the second dimension factor. The sewn collar partition is

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)} = K_t(1).$$

If a distinct large-edge branch has the controlled expansion

$$\log Z = \sum_g N_{\text{edge}}^{2-2g} F_g$$

with uniform remainder bounds, then $g_s = N_{\text{edge}}^{-1}$ rewrites the coefficient as

$$N_{\text{edge}}^{2-2g} = g_s^{2g-2},$$

the standard closed-string genus weight. $\square$

Theorem 7.2 (Edge-sum universality theorem). *On the controlled large-N_{edge} branch,*

$$\log Z_{\text{edge}} = \sum_g N_{\text{edge}}^{2-2g} F_g, \qquad g_s = \frac{1}{N_{\text{edge}}}.$$

The same representation sum gives the two-dimensional Yang-Mills heat-kernel basis and the world-sheet string basis.

Proof. The representation sum is the heat-kernel form of two-dimensional Yang-Mills. Its sewing law is the Chapman-Kolmogorov law for heat kernels, which is also the OPH collar-sewing law. The large-N_{edge} expansion reorganizes the same sewn surfaces by Euler characteristic:

$$N_{\text{edge}}^{2-2g} = g_s^{2g-2}.$$

The same edge representation sum has a 2D Yang-Mills heat-kernel reading and a perturbative closed-string worldsheet reading, provided the large-edge remainder bounds hold. □

This theorem is the bridge to the string community. It also explains why DSSYK, large-N QCD, and open-string structures keep meeting in the de Sitter flat-space limit. A 2025 DSSYK/QCD paper states that double-scaled SYK at infinite temperature and large-N QCD have large-N expansions of the same form, and that DSSYK provides a tractable window into the fixed-coupling flat-space limit [17]. A companion DSSYK flat-space paper connects the limit to strongly coupled $(1+1)$-dimensional QCD and open-string Regge behavior [18]. OPH reads this family of coincidences as different bases for sewn edge-sector sums.

8 What an OPH String Is

The heat-kernel identity gives the partition-function trace of a string. The object-level dictionary is finite. In OPH, a string is a stable one-dimensional edge-cycle normal form in the overlap carrier. A continuum string is the scaling description of that cycle after collar sewing and large-edge coarse graining.

Definition 8.1 (Fixed-cutoff OPH pre-string). *Let*

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}, \{\mathcal{I}_e\}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\})$$

be a fixed OPH patch carrier. A fixed-cutoff pre-string is a cyclic collar word

$$\Gamma = (e_1, e_2, \dots, e_n; e_{n+1} = e_1)$$

in the overlap nerve, together with edge labels and vertex intertwiners

$$\mathcal{S}_0(\Gamma) = \left(\Gamma, \{R_{e_a}\}_{a=1}^n, \{I_{v_a}\}_{a=1}^n, \{r_{e_a}\}_{a=1}^n\right),$$

where each R_e is a compact-gauge representation label carried by the edge-center algebra, each

$$I_v \in \mathrm{Hom}_G(R_{e_{a-1}} \otimes R_{e_a}, \mathbf{1} \oplus \cdots)$$

is the local fusion/sewing datum at a patch junction, and r_e is the record state exposed on the collar.

Definition 8.2 (OPH string)**.** *An OPH string is the quotient-normal-form class*

$$\mathcal{S} = [\mathcal{S}_0(\Gamma)]_{\mathrm{nf}}/\big(\textit{local repair}, \textit{collar refinement}, \textit{OPH-invisible relabeling}\big).$$

A closed string is a cyclic class with no external endpoint. An open string is an interval-class whose endpoints lie on declared observer branes, defect collars, or boundary record sectors. A multi-string state is a finite disjoint union of such classes, modulo the same repair quotient.

The cycle is visible through representation data and records on overlaps. Its support is one-dimensional in the overlap nerve. Its physical role comes from survival under local repair and refinement.

8.1 One string, many strings, and worldsheets

A single OPH string at one checkpoint is

$$\mathcal{S}(\tau_0) = [\Gamma(\tau_0), R(\tau_0), I(\tau_0), r(\tau_0)]_{\mathrm{nf}}.$$

A history of the same object through accepted repairs is a sequence

$$\mathcal{S}(\tau_0) \to \mathcal{S}(\tau_1) \to \cdots \to \mathcal{S}(\tau_m).$$

The two-dimensional worldsheet is the swept collar complex

$$\Sigma_{\mathcal{S}} := \bigcup_{j=0}^{m-1} \Gamma(\tau_j) \times [\tau_j, \tau_{j+1}],$$

with sewing at repair events. Pair-of-pants splitting and joining are OPH cobordisms in which one cyclic normal form changes into two, or two change into one, preserving exposed overlap constraints.

The dictionary is:

$$\begin{array}{ccl} \text{OPH cyclic edge normal form} & \longleftrightarrow & \text{string at one time,} \\ \text{accepted repair history of the cycle} & \longleftrightarrow & \text{worldsheet,} \\ \text{cycle split/merge cobordism} & \longleftrightarrow & \text{string interaction,} \\ \text{finite edge capacity} & \longleftrightarrow & g_s^{-1} \text{ on the controlled large-edge branch.} \end{array}$$

8.2 Why the string vibrates

The string vibrates because the cyclic edge normal form has internal normal modes. Linearize the repair law around a stable cyclic solution $\mathcal{S}_\star$. If ξ_a denotes a small edge-position or edge-label perturbation at the a-th collar site, the quadratic mismatch is

$$\Phi(\mathcal{S}_\star + \xi) = \Phi(\mathcal{S}_\star) + \frac{1}{2}\sum_{a,b} \xi_a K_{ab} \xi_b + O(\xi^3).$$

On a long uniform cycle, the Hessian K is a graph Laplacian plus local mass/curvature terms. The normal modes are Fourier modes

$$\xi_a^\mu(\tau) = \sum_{n\in\mathbb{Z}} \xi_n^\mu(\tau) e^{2\pi i n a/N}.$$

In the continuum edge limit, with $a/N \to \sigma/(2\pi)$, this becomes the closed-string field

$$X^\mu(\sigma,\tau) = x_0^\mu + p^\mu\tau + i\sqrt{\frac{\alpha'_{\mathrm{OPH}}}{2}}\sum_{n\neq 0}\frac{1}{n}\left(\alpha_n^\mu e^{-in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-in(\tau+\sigma)}\right).$$

The left/right split is the two-sided collar split. One side of the sewn edge supplies the left movers and the other supplies the right movers. Quantization of the finite symplectic record/edge normal modes gives the oscillator algebra in the scaling limit,

$$[\alpha_m^\mu, \alpha_n^\nu] = m\delta_{m+n,0}\eta^{\mu\nu}, \qquad [\tilde{\alpha}_m^\mu, \tilde{\alpha}_n^\nu] = m\delta_{m+n,0}\eta^{\mu\nu}.$$

At this stage the range of μ is not fixed by the heat-kernel theorem or by the oscillator form alone. The critical dimension and the heterotic internal central charge require the critical-edge certificate in Section 9. The parameter α'_{OPH} is the continuum normalization of the edge diffusion/repair time. Its identification with a physical string tension is conditional on matching the four-dimensional Newton constant, the OPH pixel scale, and a certified compactification threshold.

Theorem 8.3 (String emergence theorem)**.** *Assume a fixed-cutoff compact edge branch, collar sewing, a separated refinement system, and a controlled large-N_{edge} limit in which cyclic normal forms have a continuum embedding $X : \Sigma \to M_{\mathrm{OPH}}$. Then the OPH edge-cycle normal-form category maps to a perturbative string worldsheet category. The map sends cyclic edge normal forms to string states, accepted repair histories to worldsheets, and edge-cycle split/merge cobordisms to string interactions.*

Proof. At fixed cutoff, the object is finite: a cyclic word of edge collars with compact representation labels and intertwiners. Sewing gives the heat-kernel partition described above. Refinement turns long cyclic words into one-dimensional continua. A time-ordered repair history sweeps a cellulated two-complex. The large-edge expansion supplies the genus weighting. The normal-mode expansion of the repaired cycle gives the oscillator degrees of freedom. These are precisely the data used by a perturbative worldsheet description. □

Remark 8.4 (Claim boundary). The exact fixed-cutoff theorem is the edge heat-kernel/sewing theorem. The continuum string, oscillator algebra, and genus expansion require the large-edge/refinement hypotheses. OPH begins with finite edge data; string theory is the controlled language of its sewn-edge scaling branch.

9 Critical Central-Charge Gate

The OPH screen and particle papers supply the relevant integer targets for a heterotic branch [4, 5]. Those integers become central charges only after the gate below. The screen-microphysics construction gives twelve primitive curvature/record ports,

$$n_{\mathrm{port}} = 12,$$

and the reversible write/verify register gives the oriented count

$$m_{\mathrm{rep}} = 2\dim\big(\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)\big) = 2(8+3+1) = 24.$$

These are finite-register counts. A worldsheet central charge counts gapless conformal degrees of freedom, weighted by field type. For example, twenty-four chiral bosons have $c = 24$, but twenty-four chiral Majorana fermions have $c = 12$, and massive fields do not contribute to the infrared central charge [19, 20].

The heat-kernel identity also cannot determine c_L and c_R by itself. One can tensor the sewn edge sector with a spectator CFT without changing

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)} = K_t(1),$$

while shifting the total central charge. Therefore port counting and the heat-kernel trace feed a criticality test. Criticality requires the finite-carrier critical-edge certificate below.

Theorem 9.1 (Boundary from OPH edge data)**.** *The OPH edge statements used here do not imply*

$$\operatorname{rank} k_L = 24, \qquad \operatorname{rank} k_R = 12, \qquad T_{\text{OPH}} = T_{\text{Sug}}, \qquad G^2 \sim T_R.$$

Indeed the rank equations are not invariantly meaningful until a continuum current algebra and its level matrices k_L, k_R have been specified.

Proof. There are four independent obstructions.

First, the twelve curvature ports are not themselves twelve independent currents. If one defines defect fluctuations q_a, $a = 1, \ldots, 12$, from the unit curvature defects, their total charge is fixed:

$$\sum_{a=1}^{12} q_a = 12.$$

For currents $J_a = \partial q_a$, this gives

$$\sum_{a=1}^{12} J_a = 0, \qquad \sum_a k_{ab} = 0,$$

so the all-ones vector lies in the kernel of the current two-point matrix and

$$\operatorname{rank} k \le 11.$$

On the exact unit-defect normal form, $q_a = 1$ for every port, these curvature currents vanish. Thus the curvature-defect variables cannot be the required twelve independent chiral fields; any such fields must be new dynamical variables living at the ports.

Second, orientation labels do not by themselves double the rank. If verification is the reverse of writing, $J_{a,-} = -J_{a,+}$, and the unoriented covariance is K, then

$$K_{\text{orient}} = \begin{pmatrix} K & -K \\ -K & K \end{pmatrix} \sim \begin{pmatrix} 0 & 0 \\ 0 & 2K \end{pmatrix}.$$

Hence

$$\operatorname{rank} K_{\text{orient}} = \operatorname{rank} K,$$

not twice that rank. Doubling would require an independent covariance block

$$K_{\text{independent}} = \begin{pmatrix} K & 0 \\ 0 & K \end{pmatrix}.$$

Both covariance structures are compatible with two operation labels. OPH counts write/verify operations; independent-field covariance is a separate certificate.

Third, twelve Standard Model generators are not automatically central charge 12. For a non-abelian affine algebra $\widehat{\mathfrak{g}}_k$, the Sugawara central charge is

$$c(\mathfrak{g}_k) = \frac{k\ \dim \mathfrak{g}}{k + h^\vee}$$

[20]. Interpreting the Standard Model algebra as level-one affine currents gives

$$\mathfrak{g}_{\mathrm{SM}} = \mathfrak{su}(3)_1 \oplus \mathfrak{su}(2)_1 \oplus \mathfrak{u}(1),$$

and hence

$$c = \frac{8}{1+3} + \frac{3}{1+2} + 1 = 2 + 1 + 1 = 4,$$

not 12. Two independent copies would give $c = 8$, not 24. To obtain $c = 12$ or $c = 24$, the edge slots must be proved equivalent to independent Heisenberg/free-boson currents, or to another CFT with those central charges.

Fourth, a current algebra can be a proper subsector of a larger CFT:

$$T_{\mathrm{OPH}} = T_{\mathrm{Sug}} + T_{\mathrm{coset}}, \qquad c_{\mathrm{OPH}} = c_{\mathrm{Sug}} + c_{\mathrm{coset}}.$$

The equality $T_{\mathrm{OPH}} = T_{\mathrm{Sug}}$ requires $T_{\mathrm{coset}} = 0$. Neither the twelve-port theorem nor $Z_{\mathrm{edge}} = K_t(1)$ proves this. Likewise, a worldsheet supercurrent must be a local, fermion-odd field of conformal weight 3/2 satisfying the super-Virasoro OPE. A square root of a discrete repair operation is not enough; one needs a $\mathbb{Z}_2$ grading, fermionic locality, spin structures, and the supercurrent OPE. □

The precise claim is conditional: a heterotic edge-CFT completion would make the ranks, stress-tensor equality, supercurrent, and critical dimension follow rigidly.

Definition 9.2 (Critical-edge finite-carrier receipts)**.** *Let V_{port} be the real vector space of dynamical port-amplitude perturbations, distinct from the fixed curvature charges $q_a = 1$, and let*

$$\mathfrak{g}_{\mathrm{SM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \qquad \dim \mathfrak{g}_{\mathrm{SM}} = 12.$$

The finite-carrier critical-edge certificate consists of five receipts.

1. *A visible-response map*

   $$\mathcal{R}_{\mathrm{port}} : \mathfrak{g}_{\mathrm{SM}} \to V_{\mathrm{port}}$$

   with

   $$\operatorname{rank} \mathcal{R}_{\mathrm{port}} = 12.$$

2. *Positive covariance levels in every icosahedral and oriented sector:*

   $$\kappa_{\mathbf{1}}, \kappa_{\mathbf{3}}, \kappa_{\mathbf{3}'}, \kappa_{\mathbf{5}} > 0, \qquad \kappa_{\rho,\pm} > 0$$

   for $\rho \in \{\mathbf{1}, \mathbf{3}, \mathbf{3}', \mathbf{5}\}$.

3. *Scaled additive edge records whose continuum OPE is abelian Heisenberg:*

   $$J_A(z) J_B(w) \sim \frac{k_{AB}}{(z-w)^2}.$$

4. A local graded right-moving half-repair operator Q_N such that

$$\{Q_N, (-1)^F\} = 0, \qquad Q_N^2 - H_{R,N} \to 0$$

in the refinement limit.

5. A torus and spin-structure receipt

$$Z_{\mathrm{OPH}}(\tau, \bar{\tau}) = q^{-1}\bar{q}^{-1/2}(1 + \cdots)$$

with the modular transformations of the heterotic spin-structure sum.

9.1 Operational finite-carrier receipt protocol

The finite-carrier certificate is an executable protocol, not a synonym for a single twelve-port cavity. A single Echosahedron is a recurrent zero-spatial-dimensional patch: it evolves and self-reads, but the spatial edge coordinate needed for a (1+1)-dimensional scaling theory is absent. Central charge, current modes, and Virasoro closure require a sewn edge cylinder

$$\mathrm{EDGE\_CYLINDER}(L), \qquad \mathcal{H}_L = \bigotimes_{x=0}^{L-1} \mathcal{H}_{\mathrm{cell},x}, \qquad x \equiv x + L.$$

The local update on this cylinder must be inherited from the OPH read–compare–repair–commit cycle, not chosen to reproduce a desired CFT. Its exported object is a transfer operator

$$T_L = e^{-aH_L},$$

or, for a stochastic carrier, a Markov/Koopman transfer operator with a reflected-positive Euclidean interpretation.

The operational receipt suite is:

FC-1. Carrier and refinement manifest. Export model hashes, local factor dimensions, update/repair gates, transfer callbacks, A_5 action, orientation involution, translation operator, record/port operators, refinement maps, precision model, and either a positive self-adjoint transfer matrix or reflected-positive Gram matrices.

FC-2. Twelve-port dynamical rank. Perturb the twelve ports around settled states, measure the full response matrix, decompose the A_5 representation $\mathbb{R}^{12} \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$, and require all irreducible response coefficients to remain nonzero under refinement.

FC-3. Orientation independence. Form $V_{\mathrm{port}} \otimes \mathbb{R}[C_2]$, project to the $A_5 \times C_2$ sectors, and verify that both orientation parities survive dynamically. This is the immediate falsifier for mere write/verify duplication.

FC-4. Conserved currents. Discover local densities and currents satisfying the discrete continuity equation on the edge cylinder, then measure chiral level matrices whose stable ranks approach

$$\mathrm{rank}\, k_L = 24, \qquad \mathrm{rank}\, k_R = 12.$$

FC-5. Virasoro central charge. Construct lattice stress-tensor modes and fit the Virasoro residuals on low-energy projectors, with independent finite-size energy checks, targeting

$$c_L \to 24, \qquad c_R \to 12.$$

FC-6. Sugawara exhaustion. Build the Sugawara stress tensor from the measured currents and require $T_{\mathrm{OPH}} - T_{\mathrm{Sug}} \to 0$, equivalently no residual positive-central-charge coset.

FC-7. Supercurrent. Exhibit a genuine right-moving graded sector and a local odd supercurrent whose finite-carrier square-root receipt satisfies $Q_N^2 - H_{R,N} \to 0$ on the low-energy projector.

FC-8. Torus, spin structures, and internal lattice. Export spin-structure partition functions, verify their modular transformation law after the full GSO projection, and identify the rank-sixteen even self-dual left-moving internal lattice. The Bouchard-Donagi witness belongs to the $E_8 \oplus E_8$ heterotic branch.

FC-9. Blind refinement and falsification. Pre-register several (L, a) refinements, report all failures, and run controls that collapse orientation, permute ports, break icosahedral geometry, add spectators or masses, reverse chirality conventions, and disconnect seams.

The $E_8/\mathrm{Spin}(8)$ triality sidecar is useful only as a local algebraic witness for this last internal-lattice lane. It gives exact finite matrix evidence that one E_8 lattice can carry triality-related vector and positive-half-spin $2{\cdot}\mathrm{Alt}(9)$ presentations with distinct mod-2 orbit fingerprints. It is not a rank-sixteen $E_8 \oplus E_8$ certificate, not a spin-structure or supercurrent receipt, and not a replacement for the heterotic edge-polarization gate above.

The existing twelve-port calibration model can test FC-0, FC-1, and the precursor part of FC-2. It cannot by itself test rank $k_L = 24$, rank $k_R = 12$, $T_{\mathrm{OPH}} = T_{\mathrm{Sug}}$, the right-moving supercurrent, or $(c_L, c_R) = (24, 12)$. Those require the edge-cylinder extension. Without such receipts, the heterotic central-charge identification is not certified, not a passed numerical result.

Lemma 9.3 (Port-spectrum and icosahedral rank tests)**.** *If every nonzero observer-visible repair generator changes some dynamical port record, then $\mathcal{R}_{\mathrm{port}}$ is an isomorphism. If the twelve ports transform as the vertex permutation representation of A_5, then*

$$\mathbb{R}^{12} \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

For an A_5-invariant right-moving covariance matrix,

$$k_R = \kappa_{\mathbf{1}} P_{\mathbf{1}} + \kappa_{\mathbf{3}} P_{\mathbf{3}} + \kappa_{\mathbf{3}'} P_{\mathbf{3}'} + \kappa_{\mathbf{5}} P_{\mathbf{5}},$$

and

$$\operatorname{rank} k_R = 12 \quad \Longleftrightarrow \quad \kappa_{\mathbf{1}}, \kappa_{\mathbf{3}}, \kappa_{\mathbf{3}'}, \kappa_{\mathbf{5}} > 0.$$

For the oriented space $V_{\mathrm{port}} \otimes \mathbb{R}[C_2]$,

$$\operatorname{rank} k_L = 24 \quad \Longleftrightarrow \quad \kappa_{\rho,\pm} > 0 \quad \textit{for all eight oriented sectors.}$$

Proof. The response-map condition says $\ker \mathcal{R}_{\mathrm{port}} = 0$. Since the domain and codomain both have dimension 12, $\mathcal{R}_{\mathrm{port}}$ is invertible. The A_5 permutation character on the twelve icosahedral vertices is

$$\chi_V = (12, 0, 0, 2, 2),$$

which decomposes as

$$\chi_{\mathbf{1}} + \chi_{\mathbf{3}} + \chi_{\mathbf{3}'} + \chi_{\mathbf{5}} = (12, 0, 0, 2, 2).$$

Schur's lemma then diagonalizes any A_5-invariant covariance by irreducible sector. Reflection positivity makes nonzero sector coefficients positive. The oriented statement is the same argument applied to the $A_5 \times C_2$ decomposition. □

Lemma 9.4 (Observability and current emergence)**.** *If the current covariance is positive semidefinite, all invisible directions have been quotiented out, and every remaining nonzero port perturbation changes an observer-visible record statistic, then the current covariance is positive definite. If additive port records satisfy*

$$Q_A(I_1 \cup I_2) = Q_A(I_1) + Q_A(I_2)$$

and are conserved under accepted repairs away from interval endpoints, then in a local unitary scale-invariant sewn-edge limit they become chiral currents

$$Q_A(I) = \int_I J_A(z)\, dz, \qquad \bar{\partial} J_A = 0.$$

If the fixed-cutoff integrated charges commute, their continuum current algebra is abelian:

$$J_A(z) J_B(w) \sim \frac{k_{AB}}{(z-w)^2}.$$

Proof. For a positive-semidefinite covariance, a nonzero vector v with $v^T k v = 0$ defines a zero-norm combination $J_v = v^A J_A$, hence an invisible fluctuation in the observer-facing algebra. The quotient-observability hypotheses exclude such a vector. Additivity gives a local continuum density, conservation gives a conserved two-dimensional current, and chirality gives a holomorphic component of weight $(1, 0)$. Commuting zero modes exclude a simple-pole affine term, leaving only the central $(z - w)^{-2}$ singularity. □

Theorem 9.5 (Critical-edge certificate theorem)**.** *If the five finite-carrier receipts above hold, then the OPH sewn-edge scaling branch satisfies the heterotic edge-polarization hypothesis below. Consequently the physical edge CFT has*

$$(c_L^{\text{phys}}, c_R^{\text{phys}}) = (24, 12),$$

a right-moving $N = 1$ supercurrent, no residual coset, and an even self-dual rank-sixteen left-moving internal lattice.

Proof. The rank and observability receipts make the right-moving port covariance positive definite of rank 12 and the oriented left-moving covariance positive definite of rank 24. The current-emergence receipt turns the additive records into abelian Heisenberg currents. The abelian Sugawara construction gives central charge equal to rank, hence $c_R = 12$ and $c_L = 24$ for the generated sectors.

The torus vacuum exponent

$$q^{-1}\bar{q}^{-1/2} = q^{-c_L/24}\bar{q}^{-c_R/24}$$

fixes the total central charges to the same values. Therefore the residual stress tensor

$$T_{\text{res}} = T_{\text{OPH}} - T_{\text{Sug}}$$

has $c_{\text{res}} = 0$ in both chiralities. In a unitary positive-energy CFT with unique vacuum, a $c = 0$ Virasoro sector is null after quotienting null states, so there is no residual coset.

The local graded half-repair operator supplies the refinement limit of a right-moving odd square root of translation. Its density is the supercurrent G_R, and the free $N = 1$ OPE gives

$$G_R(\bar{z}) G_R(\bar{w}) \sim \frac{2c_R/3}{(\bar{z} - \bar{w})^3} + \frac{2T_R(\bar{w})}{\bar{z} - \bar{w}}.$$

The modular spin-structure receipt then identifies the sixteen remaining left-moving units with an even self-dual rank-sixteen lattice sector. □

Assumption 9.6 (Heterotic edge-polarization hypothesis)**.** *The physical sewn-edge scaling limit is isomorphic, as a graded chiral CFT, to*

$$\mathrm{ScaleLim}(\textit{OPH sewn-edge carrier}) \cong (\mathrm{Heis}_8 \otimes V_{\Gamma_{16}})_L \otimes (\mathrm{Heis}_8 \otimes \mathrm{Ferm}_8)_R\,,$$

with no residual coset or spectator CFT. Here Heis_8 *is generated by eight transverse free bosons,* $V_{\Gamma_{16}}$ *is a rank-sixteen even self-dual lattice VOA on the left-moving side, and* Ferm_8 *is generated by eight right-moving Majorana fermions with a consistent spin-structure projection.*

Theorem 9.7 (Heterotic completion theorem)**.** *Under Assumption 9.6, the physical chiral central charges are*

$$\boxed{c_L^{\mathrm{phys}} = 24, \qquad c_R^{\mathrm{phys}} = 12,}$$

the relevant Heisenberg level matrices have ranks 24 *and* 12*, the stress tensors are Sugawara stress tensors for those Heisenberg currents, and the right-moving sector carries an* $N = 1$ *supercurrent. The shared target spacetime dimension is then forced to be* $D = 10$*, with left-moving internal central charge* 16*.*

Proof. On the left, write

$$\Phi_L^A = (X_L^1, \dots, X_L^8, Y_L^1, \dots, Y_L^{16}), \qquad A = 1, \dots, 24,$$

with

$$\Phi_L^A(z)\Phi_L^B(w) \sim -\delta^{AB}\log(z-w).$$

The currents $J_L^A = i\partial\Phi_L^A$ obey

$$J_L^A(z)J_L^B(w) \sim \frac{\delta^{AB}}{(z-w)^2},$$

so $k_L = I_{24}$ and $\operatorname{rank} k_L = 24$. The abelian Sugawara tensor

$$T_L = \frac{1}{2}\sum_{A=1}^{24} : J_L^A J_L^A :$$

has Virasoro central charge $c_L = 24$, and by the no-coset clause it is the full left-moving stress tensor.

On the right, take eight transverse bosons X_R^i and eight Majorana fermions ψ^i:

$$X_R^i(\bar z)X_R^j(\bar w) \sim -\delta^{ij}\log(\bar z - \bar w), \qquad \psi^i(\bar z)\psi^j(\bar w) \sim \frac{\delta^{ij}}{\bar z - \bar w}.$$

Pairwise bosonization of the fermions,

$$\frac{\psi^{2a-1} \pm i\psi^{2a}}{\sqrt{2}} = e^{\pm iH^a}, \qquad a = 1, \dots, 4,$$

shows that their stress tensor is equivalent to four chiral bosons. A Heisenberg-current basis is

$$J_R^\alpha = \left(i\bar\partial X_R^1, \dots, i\bar\partial X_R^8, i\bar\partial H^1, \dots, i\bar\partial H^4\right), \qquad \alpha = 1, \dots, 12,$$

with level matrix $k_R = I_{12}$. Thus $\operatorname{rank} k_R = 12$, and the right-moving stress tensor is the full abelian Sugawara tensor in the bosonized basis:

$$T_R = \frac{1}{2}\sum_{\alpha=1}^{12} : J_R^\alpha J_R^\alpha :, \qquad c_R = 8 + \frac{8}{2} = 12.$$

The unbosonized expression gives the standard $N = 1$ supercurrent

$$G_R(\bar{z}) = i \sum_{i=1}^{8} \psi^i(\bar{z})\bar{\partial} X_R^i(\bar{z}).$$

Wick contraction yields

$$G_R(\bar{z})G_R(\bar{w}) \sim \frac{8}{(\bar{z}-\bar{w})^3} + \frac{2T_R(\bar{w})}{\bar{z}-\bar{w}}.$$

Since $2c_R/3 = 2(12)/3 = 8$, this is precisely the $N = 1$ super-Virasoro OPE, so

$$G^2 \sim T_R.$$

Let $n = D - 2$ be the number of physical transverse spacetime coordinates. On the supersymmetric chirality,

$$c_R^{\text{phys}} = n + \frac{n}{2} = \frac{3}{2}n.$$

Setting $c_R^{\text{phys}} = 12$ gives $n = 8$, hence

$$\boxed{D = 10.}$$

On the bosonic chirality,

$$c_L^{\text{phys}} = n + c_{\text{internal}} = 24,$$

so

$$\boxed{c_{\text{internal}} = 16.}$$

Covariantly,

$$c_L^{\text{matter}} = 10 + 16 = 26, \qquad c_R^{\text{matter}} = 10 + \frac{10}{2} = 15,$$

the matter central charges canceled by the bosonic ghosts on the left and the supersymmetric ghost system on the right. □

Theorem 9.8 (Modular lattice and anomaly closure)**.** *The rank-sixteen internal lattice in Assumption 9.6 must be even and self-dual. Hence*

$$\Gamma_{16} = E_8 \oplus E_8 \quad \textit{or} \quad \Gamma_{16} = D_{16}^+,$$

corresponding to the heterotic gauge groups $E_8 \times E_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$. The Bouchard-Donagi selector used in this paper chooses the $E_8 \oplus E_8$ branch. With $D = 10$, the covariant Weyl anomalies cancel:

$$(10 + 16) - 26 = 0, \qquad \left(10 + \frac{10}{2}\right) - 15 = 0.$$

Proof. For the internal bosons,

$$Z_{\Gamma_{16}}(\tau) = \frac{\Theta_{\Gamma_{16}}(\tau)}{\eta(\tau)^{16}}, \qquad \Theta_\Gamma(\tau) = \sum_{p \in \Gamma} q^{p^2/2}.$$

Invariance under $T : \tau \mapsto \tau + 1$ requires $p^2 \in 2\mathbb{Z}$, so the lattice is even. Under $S : \tau \mapsto -1/\tau$, Poisson resummation maps Γ to its dual Γ^*, so modular closure requires $\Gamma^* = \Gamma$. The positive-definite even unimodular rank-sixteen lattices are $E_8 \oplus E_8$ and D_{16}^+. The ghost central charges are -26 on the bosonic left and -15 on the supersymmetric right, giving the displayed anomaly cancellations. BRST nilpotency follows with the standard intercept and level-matching conditions. □

Remark 9.9 (Local E_8 triality receipt)*.* The finite $E_8/\mathrm{Spin}(8)$ triality certificate can be used as a sanity check for the E_8-side automorphism and representation bookkeeping in the $E_8 \oplus E_8$ branch. It constructs a nonsplit $2{\cdot}\mathrm{Alt}(9)$ subgroup whose positive-half-spin image preserves an E_8 lattice and whose vector and spin-side mod-2 orbit fingerprints are not conjugate before triality fusion. This is a local E_8 subgroup certificate. It does not identify the full rank-sixteen internal lattice, prove modular invariance, supply the GSO/spin-structure sum, or close the OPH critical-edge CFT gate.

The OPH theorem target is precise:

$$\boxed{\mathrm{ScaleLim}(\text{OPH sewn-edge carrier}) \cong (\mathrm{Heis}_8 \otimes V_{\Gamma_{16}})_L \otimes (\mathrm{Heis}_8 \otimes \mathrm{Ferm}_8)_R}$$

with no residual coset. It cannot be replaced by the numerical observation $12 \to 24$. It must be established by constructing the edge transfer operator, identifying the gapless chiral fields, calculating their OPEs, deriving the fermionic grading and spin-structure sum, and proving stress exhaustion. Equivalently, the finite carrier must export the five receipts listed above:

$$\operatorname{rank} \mathcal{R}_{\mathrm{port}} = 12, \quad \kappa_\rho > 0, \quad \kappa_{\rho,\pm} > 0, \quad J_A J_B \sim k_{AB}(z-w)^{-2}, \quad Q_N^2 - H_{R,N} \to 0,$$

together with the torus receipt $Z_{\mathrm{OPH}} = q^{-1}\bar{q}^{-1/2}(1+\cdots)$ and the correct modular spin-structure transformations. Once that edge-VOA identification is proved, $D = 10$ and the left-moving $c_{\mathrm{internal}} = 16$ sector follow by the calculation above. The number 26 belongs to the left-moving bosonic matter central charge, not to twenty-six shared heterotic spacetime dimensions.

10 The OPH Graviton and the String Graviton

The graviton dictionary is the conceptual hinge. The string graviton is the left-right edge-mode factorization of the OPH graviton. OPH identifies its metric quantum with the ordinary massless closed-string spin-two state.

10.1 OPH side: metric from modular geometry

On the OPH gravity branch, cap modular flow becomes geometric in the support-visible scaling limit, and fixed-cap generalized-entropy stationarity supplies the Einstein relation. The metric is the compressed observer-facing datum that makes the cap modular action geometric. A small metric perturbation is a deformation of the OPH geometric normal form:

$$q_{ab}^{\mathrm{OPH}} \longmapsto q_{ab}^{\mathrm{OPH}} + h_{ab}^{\mathrm{OPH}}.$$

The transverse-traceless part

$$h_{ab}^{\mathrm{OPH,TT}}$$

is the propagating spin-two perturbation. Quantizing this field gives the OPH graviton.

10.2 String side: metric from the closed-string vertex

In a critical closed string, the metric appears as a background coupling in the sigma model:

$$S_\sigma \supset \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{\gamma}\, \gamma^{\alpha\beta} G_{\mu\nu}(X) \partial_\alpha X^\mu \partial_\beta X^\nu.$$

Perturbing

$$G_{\mu\nu} = G^{(0)}_{\mu\nu} + \kappa h_{\mu\nu}$$

produces the standard closed-string spin-two vertex

$$V_h(k,\epsilon) = \epsilon_{\mu\nu}\,\partial X^\mu \bar{\partial} X^\nu e^{ik\cdot X}.$$

Equivalently, at the first closed-string oscillator level,

$$|G;k,\epsilon\rangle = \epsilon^{\mathrm{TT}}_{\mu\nu}\alpha^\mu_{-1}\tilde{\alpha}^\nu_{-1}|0;k\rangle.$$

The polarization decomposes into symmetric traceless, antisymmetric, and trace pieces:

$$\epsilon_{\mu\nu} = \epsilon^{\mathrm{TT}}_{(\mu\nu)} + \epsilon_{[\mu\nu]} + \frac{1}{D}\eta_{\mu\nu}\epsilon^\lambda{}_\lambda.$$

The symmetric transverse-traceless piece is the string graviton. The antisymmetric piece is the B-field, and the trace is the dilaton.

10.3 Left-right collar factorization

A closed OPH edge cycle has two collar readings. The two readings are sewn into the heat-kernel factor d_R^2. In the large-edge worldsheet language they become the left and right chiral oscillator sectors:

$$\text{left collar edge mode} \otimes \text{right collar edge mode} \longmapsto \alpha^\mu_{-1}\tilde{\alpha}^\nu_{-1}|0;k\rangle.$$

The level-one tensor product splits after contraction with a polarization tensor. A general polarized state decomposes as

$$\epsilon_{\mu\nu}\alpha^\mu_{-1}\tilde{\alpha}^\nu_{-1}|0;k\rangle = \epsilon^{\mathrm{TT}}_{(\mu\nu)}\alpha^\mu_{-1}\tilde{\alpha}^\nu_{-1}|0;k\rangle + \epsilon_{[\mu\nu]}\alpha^\mu_{-1}\tilde{\alpha}^\nu_{-1}|0;k\rangle + \frac{1}{D}\eta_{\mu\nu}\epsilon^\lambda{}_\lambda\alpha^\mu_{-1}\tilde{\alpha}^\nu_{-1}|0;k\rangle.$$

The OPH graviton is the symmetric transverse-traceless component of the two-sided collar excitation read in the critical worldsheet language:

$$h^{\mathrm{OPH,TT}}_{\mu\nu} \longleftrightarrow \epsilon^{\mathrm{TT}}_{\mu\nu}\alpha^\mu_{-1}\tilde{\alpha}^\nu_{-1}|0;k\rangle.$$

10.4 The map

Define the OPH-to-string graviton map by

$$\mathfrak{M}_{\mathrm{grav}} : [h^{\mathrm{OPH,TT}}_{ab}] \longmapsto [\epsilon^{\mathrm{TT}}_{\mu\nu}\partial X^\mu \bar{\partial} X^\nu e^{ik\cdot X}],$$

with the following normalization condition:

$$G_4^{\mathrm{string}}(m_\star, \alpha', g_s, V_6) = G_4^{\mathrm{OPH}}(N_{\mathrm{scr}}, P).$$

OPH fixes the Einstein-frame graviton normalization, and the string representative supplies the critical worldsheet realization.

Theorem 10.1 (Graviton identification theorem)**.** *On an OPH-correct critical-string presentation, the massless closed-string symmetric spin-two state is exactly the quantized OPH metric perturbation:*

$$\mathfrak{M}_{\mathrm{grav}}(h^{\mathrm{OPH,TT}}) = h^{\mathrm{string,TT}}.$$

No additional independent massless spin-two field is allowed in the OPH-visible sector.

Proof. The OPH gravity branch produces a single observer-facing Einstein-frame metric on the physical quotient. Its propagating transverse-traceless perturbation is a massless spin-two field. A critical closed string contains a massless symmetric spin-two state, represented by the vertex $\epsilon_{\mu\nu}\partial X^\mu \bar{\partial} X^\nu e^{ik\cdot X}$. In an OPH-correct lift, the worldsheet background metric is the same observer-facing metric that OPH reconstructs. Its symmetric transverse-traceless vertex is the quantization of the OPH metric perturbation. If an extra independent massless spin-two field survived in the same visible sector, the four-dimensional low-energy theory would contain an additional graviton or bigravity degree of freedom, changing the OPH Einstein branch and failing OPH-equivalence. The two gravitons are the same field. □

OPH object	**String-theory image**
Cap modular geometry	Target-space metric background $G_{\mu\nu}(X)$.
Fixed-cap generalized entropy stationarity	Low-energy Einstein equation / vanishing metric beta function on the selected background.
Metric perturbation h_{ab}^{OPH}	Closed-string background deformation $h_{\mu\nu}\partial X^\mu \bar{\partial} X^\nu$.
OPH graviton	Symmetric transverse-traceless massless closed-string state.
Screen-capacity normalization	Four-dimensional Newton constant after compactification.
Edge record/collar sewing	Worldsheet sewing and genus expansion in the controlled large-edge branch.

11 OPH-Augmented String Theory

OPH supplies a selection and readout functor for critical string vacua. Ordinary string theory supplies a large class of critical presentations. OPH asks which presentation is the critical worldsheet completion of the observer-visible edge normal form.

Definition 11.1 (OPH augmentation of a string vacuum)**.** *A string vacuum v is augmented by OPH data when it is equipped with a map*

$$\Pi_{\mathrm{OPH}}(v) = (G_{\mathrm{phys}}, Y, N_c, N_g, n_H, \mathcal{O}_{\mathrm{op}}, G_4, \Lambda, \mathcal{O}_{\mathrm{quant}})$$

from compactification data to observer-visible records, together with the constraints

$$\Pi_{\mathrm{OPH}}(v) = \Pi^{\mathrm{OPH}}_{\mathrm{target}}, \qquad \operatorname{rank} D\Pi_{\mathrm{OPH}}|_{N_{\mathrm{OPH}}} = \dim N_{\mathrm{OPH}},$$

where N_{OPH} is the moduli subspace transverse to OPH-invisible redundancies. The rank condition is the moduli-locking condition: the target has no adjustable flat direction after the OPH quotient.

The augmented vacuum equations are:

$$\begin{cases} \beta^G = \beta^B = \beta^\Phi = 0, & \text{worldsheet consistency,} \\ (c_L^{\mathrm{phys}}, c_R^{\mathrm{phys}}) = (24, 12), \quad G^2 \sim T_R, & \text{critical-edge certificate,} \\ F^{0,2} = 0, \quad J^2 \wedge F = 0, & \text{heterotic bundle stability/HYM,} \\ c_2(TX) - c_2(V_{\mathrm{vis}}) - c_2(V_{\mathrm{hid}}) = [W], & \text{Bianchi/anomaly condition,} \\ \chi(V_{\mathrm{matter}}) = 3, \quad n_H = 1, \quad N_{\mathrm{exotic}} = 0, & \text{visible cohomology target,} \\ G_{\mathrm{phys}} = (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6, & \text{global group target,} \\ \mathcal{O}_{\mathrm{op}} = \mathbb{Z}_4^R \text{ safety}, & \text{operator target,} \\ G_4^{\mathrm{string}} = G_4^{\mathrm{OPH}}, \quad \alpha' \mapsto \alpha'_{\mathrm{OPH}}, & \text{graviton normalization target,} \\ \mathcal{F}_v(m_\star) = \mathcal{O}_{\mathrm{OPH}}, \quad \operatorname{rank} D\mathcal{F}_v = \text{full}, & \text{moduli/threshold target.} \end{cases}$$

This turns landscape selection into a finite certificate problem. A conventional string vacuum can be internally consistent and fail the OPH observer-visible target map.

11.1 Testable consequences after certificate closure

Once the named witness class is fixed, the theory makes gates that generic landscape reasoning lacks:

1. **One visible massless Higgs pair at the compactification target.** Extra massless Higgs pairs fail the OPH one-Higgs electroweak branch.

2. **No massless chiral exotics in the published cohomology table.** Exotic chiral matter changes the observer-visible matter package and fails the cohomology target.

3. **No extra visible low-scale U(1).** Additional visible abelian gauge bosons change the OPH global group quotient.

4. **No massless X/Y gauge bosons in the connected visible adjoint.** The selected visible group is the product quotient. Massive Wilson-line and Kaluza–Klein modes require the separate heavy-spectrum and coupling calculation.

5. **Operator-safety pattern.** Perturbative R-parity violation, perturbative dimension-five proton decay, and a perturbative μ-term are forbidden on the operator-safe branch; Yukawas and the Weinberg neutrino operator are allowed.

6. **One visible graviton.** Any additional massless spin-two field, bimetric sector, or independent low-energy tensor polarization fails the OPH graviton-identity gate.

12 String-Community Pressure Points

The string-community problem is selection. The 2024 Standard Model review states the challenge in compactification language: reproduce the Standard Model gauge sector, chiral matter, and phenomenology from geometry and topology [27]. Heterotic M-theory stabilization work emphasizes the computational weight of dilaton, complex-structure, and Kähler moduli stabilization [28]. The philosophy literature on the string landscape frames the predictivity concern as underdetermination among many solutions compatible with observed data [29]. Exact flux-vacua work also makes the computational problem concrete: stabilized flux vacua require solving vacuum conditions with exponential corrections, although symmetry loci can turn some conditions algebraic [30].

OPH imposes the observer-visible target

$$\mathfrak{S}_{\mathrm{OPH}} = \begin{pmatrix} \dfrac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, & N_c = 3, & N_g = 3, & Y_{\mathrm{SM}}, \\ n_H = 1, & \text{no light chiral exotics}, & \mathbb{Z}_4^R \text{ safety} & \end{pmatrix}.$$

The named geometric witness in the package is

$$\boxed{BD^{\mathrm{OPH}}_{n=1} = \text{Bouchard-Donagi } E_8\times E_8 \text{ heterotic SU(5) Standard Model, one-Higgs-pair stratum.}}$$

The named operator-safe candidate is

$$\boxed{BD^{\mathrm{OPH}}_{n=1,+} = BD^{\mathrm{OPH}}_{n=1} + \mathbb{Z}_4^R.}$$

Bouchard and Donagi introduced a heterotic Standard Model with the visible massless cohomology spectrum of the MSSM, no massless visible exotics, observable $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, a Calabi–Yau threefold with $\mathbb{Z}_2$ fundamental group, and an invariant $\mathrm{SU}(5)$ bundle. Depending on moduli, the model has zero, one, or two Higgs doublet conjugate pairs; the one-pair region gives precisely the massless MSSM Higgs content [21]. This cohomology calculation determines zero modes, rather than the nonzero Dirac/Laplacian, Kaluza–Klein, winding, oscillator, Wilson-line, vectorlike, five-brane, or hidden spectrum [24].

The global group identity is central:

$$\mathrm{Cent}_{\mathrm{SU}(5)}(\mathrm{diag}(1,1,1,-1,-1)) = S(\mathrm{U}(3) \times \mathrm{U}(2)),$$

$$S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

This fixes the charge lattice, beyond Lie-algebra matching.

Theorem 12.1 (OPH vacuum sieve theorem)**.** *A critical string vacuum $\mathcal{T}$ is OPH-admissible only if*

$$\Pi_{\mathrm{IR}}^{\mathrm{OPH}}(\mathcal{T}) = \mathfrak{S}_{\mathrm{OPH}}.$$

For the package studied here, the named geometric witness is $BD_{n=1}^{\mathrm{OPH}}$, and the operator-safe candidate is $BD_{n=1,+}^{\mathrm{OPH}}$.

Theorem 12.2 (Global group locking theorem)**.** *The OPH visible group and the Bouchard-Donagi Wilson-line centralizer agree:*

$$G_{\mathrm{phys}} = S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

For $\rho(-1) = \mathrm{diag}(1,1,1,-1,-1)$,

$$\mathrm{Cent}_{\mathrm{SU}(5)}(\rho) = S(\mathrm{U}(3) \times \mathrm{U}(2)).$$

Theorem 12.3 (BD structural geometry witness theorem)**.** *The Bouchard-Donagi $\mathbb{Z}_2$-quotient* $\mathrm{SU}(5)$*-bundle construction supplies a nonempty heterotic* $\mathrm{SU}(5)$*-corridor witness whose Wilson-line centralizer is the OPH global Standard Model group and whose visible massless cohomology contains the three-generation, no-exotic, one-Higgs branch used by the OPH selector. This statement does not supply a heavy-spectrum, threshold, or supersymmetric-sector decoupling certificate.*

Proof. Bouchard-Donagi construct the $E_8 \times E_8$ heterotic model on a Calabi-Yau quotient $X = \widetilde{X}/\mathbb{Z}_2$ with an invariant $\mathrm{SU}(5)$ bundle and a $\mathbb{Z}_2$ Wilson line [21]. Their bundle is built on the cover using an extension of rank-two and rank-three pieces,

$$0 \to V_2 \to \widetilde{V}^* \to V_3 \to 0, \qquad V_i = \pi'^* W_i \otimes \pi^* L_i.$$

The resulting observable zero-mode sector has the Standard Model gauge algebra, three generations, no massless visible exotic matter, and moduli regions with 0, 1, or 2 massless Higgs doublet conjugate pairs. The OPH target chooses the one-pair region. The centralizer theorem above supplies the global quotient

$$S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6.$$

The BD geometry supplies the named heterotic witness, and OPH supplies the global quotient and Higgs-stratum selector. □

13 String Problems Reduced to OPH Gates

The table lists the string-theory gates. Exact algebraic gates can close inside this paper. Cohomology reproduction, geometric realization of the $\mathbb{Z}_4^R$ safety layer, full Yukawa computation, the nonzero-mode and hidden spectra, threshold matching, the low-energy decoupling map, and moduli locking require external certificates built from the Bouchard–Donagi geometry and its physical completion.

String-theory pressure point	**OPH result**	**Claim boundary in this paper**
Vacuum selection	Converts candidate vacua into a public sieve against $\mathfrak{S}_{\mathrm{OPH}}$.	Exact selector definition; comparative audits can expand.
Global Standard Model group	Locks the finite quotient $(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6$ beyond Lie-algebra data.	Exact group proof.
Charge lattice and anomalies	Uses the Standard Model hypercharge lattice with anomaly cancellation.	Exact rational arithmetic.
Yukawa admissibility	One-Higgs Yukawa terms are gauge invariant.	Exact charge-sum proof.
Three generations	$\mathbb{Z}_2$-cover arithmetic gives $N_g = 3$.	Exact index arithmetic, assuming the BD c_3 input.
Higgs multiplicity	Selects $n = 1$ among BD $n = 0, 1, 2$ strata.	Selector proof.
Massless unified-gauge carriers	The connected massless product-group adjoint excludes an X/Y generator. Massive BD modes remain in the heavy-spectrum gate.	Exact massless Lie-algebra proof plus external heavy-mode certificate.
String emergence	Sewn edge partition equals $K_t(1)$.	Exact Peter-Weyl proof.
Worldsheet criticality	The twelve-port and twenty-four oriented-register counts define only a heterotic central-charge target until the edge VOA is identified.	Non-implication theorem plus critical-edge certificate theorem; finite-carrier receipts not emitted here.
Moduli stabilization	Becomes $\mathcal{F}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$.	Target criterion; BD map not emitted here.
Operator safety	The MSSM $\mathbb{Z}_4^R$ layer permits Yukawas and the Weinberg operator, and forbids perturbative RPV, dimension-five proton decay, and the perturbative μ-term.	Charge algebra closes; BD geometric realization is a certificate gate.
Empirical tests	Assigns the named candidate a compact list of failure-prone low-energy and threshold proxy screens.	Reproducible target equations; no branch compatibility result.

Physicist pressure point	Closure supplied by the selector	Gate type
Landscape degeneracy	OPH replaces broad vacuum search with the finite target $\mathfrak{S}_{\mathrm{OPH}}$ and a public sieve.	Selector theorem.
Global Standard Model group	The Wilson-line centralizer lands on $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.	Exact algebra.
Three families	The $\mathbb{Z}_2$-cover index relation gives $N_g = 3$ from $\lvert c_3(\widetilde{V})\rvert = 12$.	BD input plus exact arithmetic.
Higgs multiplicity	The $n = 1$ stratum is the minimal massless-cohomology stratum under the declared zero-mode score.	Selector theorem.
Exotic matter	BD supplies the no-exotics visible massless MSSM cohomology witness.	Published zero-mode geometry certificate; heavy and vectorlike modes are outside it.
Massless unified-gauge carriers	The connected massless product-group adjoint excludes mixed $(3, 2, \pm 5/6)$ generators; no conclusion about massive carriers follows.	Exact massless Lie algebra.
Critical dimension	If the sewn-edge carrier has the heterotic polarization $(\mathrm{Heis}_8 \otimes V_{\Gamma_{16}})_L \otimes (\mathrm{Heis}_8 \otimes \mathrm{Ferm}_8)_R$, then $D = 10$ and $c_{\mathrm{internal}} = 16$.	Conditional edge-VOA gate.
RPV and μ-problem	$\mathbb{Z}_4^R$ permits Yukawas, forbids perturbative RPV and dimension-five proton decay, forbids perturbative μ, and leaves matter parity.	Exact charge algebra; compactification certificate required.
Moduli stabilization	The problem becomes $\mathcal{F}_{BD,n=1,+}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$ plus full transverse rank.	Abstract isolation criterion only; no BD numerical point or rank certificate is emitted.
Predictivity	Higgs/top, stop, and one-loop gauge-running coordinates give failure-prone proxy targets.	Reproducible target-side proxies only; no spectrum or threshold certificate is emitted.

13.1 Problem 1: vacuum selection

String theory supplies many compactifications with Standard-Model-like low-energy data. OPH adds an independent target:

$$\mathfrak{S}_{\mathrm{OPH}} = \left(\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, N_c = 3,\ N_g = 3,\ Y_{\mathrm{SM}},\ n_H = 1\right), \\ \text{no light chiral exotics}, \qquad \text{no extra visible low-scale U(1)}.$$

Theorem 13.1 (Selector reduction)**.** *Let $\mathcal{C}$ be any class of critical string compactifications with an* *observer-visible infrared projection* $\Pi_{\mathrm{IR}}^{\mathrm{OPH}}$. *Define*

$$\mathcal{C}_{\mathrm{OPH}} = \{\mathcal{T} \in \mathcal{C} : \Pi_{\mathrm{IR}}^{\mathrm{OPH}}(\mathcal{T}) = \mathfrak{S}_{\mathrm{OPH}}\}.$$

The OPH selection problem over $\mathcal{C}$ *is exactly the finite or enumerable audit of* $\mathcal{C}_{\mathrm{OPH}}$ *against the gates in* $\mathfrak{S}_{\mathrm{OPH}}$ *and the quantitative target* $\mathcal{O}_{\mathrm{OPH}}$.

Proof. The OPH target $\mathfrak{S}_{\mathrm{OPH}}$ is fixed independently of any string compactification. The projection $\Pi_{\mathrm{IR}}^{\mathrm{OPH}}$ assigns each candidate its observer-visible group, charge lattice, matter content, Higgs count, and visible low-scale gauge factors. Equality with $\mathfrak{S}_{\mathrm{OPH}}$ is a conjunction of explicit gates. A candidate passes exactly when all equalities and exclusions hold. The quantitative stage appends the equation

$$\mathcal{F}_{\mathcal{T}}(m) = \mathcal{O}_{\mathrm{OPH}}.$$

The landscape question becomes a sieve and target equation. □

13.2 Problem 2: global Standard Model group

The OPH target fixes the global group beyond the Lie algebra, at the level of the charge lattice.

Theorem 13.2 (Centralizer computation)**.** *Let*

$$\rho = \mathrm{diag}(1, 1, 1, -1, -1) \in \mathrm{SU}(5).$$

The centralizer of ρ *in* $\mathrm{SU}(5)$ *is*

$$\mathrm{Cent}_{\mathrm{SU}(5)}(\rho) = S(\mathrm{U}(3) \times \mathrm{U}(2)).$$

Proof. The $+1$ eigenspace of ρ has dimension 3, and the -1 eigenspace has dimension 2. A unitary matrix commutes with ρ exactly when it preserves these two eigenspaces. Such a matrix is block diagonal,

$$g = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \qquad A \in \mathrm{U}(3), \quad B \in \mathrm{U}(2).$$

The condition $g \in \mathrm{SU}(5)$ is $\det A \det B = 1$, precisely

$$S(\mathrm{U}(3) \times \mathrm{U}(2)).$$

□

Theorem 13.3 ($\mathbb{Z}_6$ quotient)**.** *There is an isomorphism*

$$S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

Proof. Define

$$\varphi : \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1) \to S(\mathrm{U}(3) \times \mathrm{U}(2))$$

by

$$\varphi(A, B, z) = \begin{pmatrix} z^2 A & 0 \\ 0 & z^{-3} B \end{pmatrix}.$$

The determinant is

$$\det(z^2 A) \det(z^{-3} B) = z^6 z^{-6} \det A \det B = 1,$$

so the image lies in $S(\mathrm{U}(3)\times\mathrm{U}(2))$. The kernel consists of triples with

$$z^2A = I_3, \qquad z^{-3}B = I_2.$$

$A = z^{-2}I_3$ and $B = z^3 I_2$. The conditions $A\in\mathrm{SU}(3)$ and $B\in\mathrm{SU}(2)$ give

$$z^{-6} = 1, \qquad z^6 = 1.$$

The kernel is

$$\{(z^{-2}I_3, z^3I_2, z) : z^6 = 1\} \cong \mathbb{Z}_6.$$

Surjectivity follows by decomposing any $(A', B') \in S(\mathrm{U}(3)\times\mathrm{U}(2))$ into determinant-one parts and the common $\mathrm{U}(1)$ factor. The first isomorphism theorem gives the quotient. □

13.3 Problem 3: charge lattice, anomalies, and one-Higgs Yukawas

The OPH target uses the Standard Model hypercharge lattice:

$$Q = (3,2)_{1/6}, \quad u^c = (\bar{3},1)_{-2/3}, \quad d^c = (\bar{3},1)_{1/3}, \quad L = (1,2)_{-1/2}, \quad e^c = (1,1)_1, \quad H = (1,2)_{1/2}.$$

Theorem 13.4 (One-generation anomaly cancellation)**.** *For one generation with the hypercharges above,*

$$\mathrm{SU}(3)^2\mathrm{U}(1) = 0, \quad \mathrm{SU}(2)^2\mathrm{U}(1) = 0, \quad \mathrm{grav}^2\mathrm{U}(1) = 0, \quad \mathrm{U}(1)^3 = 0.$$

Proof. Use left-handed Weyl fields. For $\mathrm{SU}(3)^2\mathrm{U}(1)$, the color-charged fields give

$$2\cdot\frac{1}{2}\cdot\frac{1}{6} + \frac{1}{2}\cdot\left(-\frac{2}{3}\right) + \frac{1}{2}\cdot\frac{1}{3} = \frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0.$$

For $\mathrm{SU}(2)^2\mathrm{U}(1)$, the weak doublets give

$$3\cdot\frac{1}{2}\cdot\frac{1}{6} + \frac{1}{2}\cdot\left(-\frac{1}{2}\right) = \frac{1}{4} - \frac{1}{4} = 0.$$

For the mixed gravitational anomaly, count dimensions:

$$6\cdot\frac{1}{6} + 3\cdot\left(-\frac{2}{3}\right) + 3\cdot\frac{1}{3} + 2\cdot\left(-\frac{1}{2}\right) + 1 = 1 - 2 + 1 - 1 + 1 = 0.$$

For the cubic anomaly,

$$\begin{aligned} &6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1^3 \\ &= \frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1 = 0. \end{aligned}$$

□

Theorem 13.5 (One-Higgs Yukawa invariants)**.** *The one-Higgs Standard Model Yukawa terms have zero hypercharge:*

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Proof. The hypercharge sums are

$$\frac{1}{6} + \frac{1}{2} - \frac{2}{3} = 0,$$
$$\frac{1}{6} - \frac{1}{2} + \frac{1}{3} = 0,$$

and

$$-\frac{1}{2} - \frac{1}{2} + 1 = 0.$$

The nonabelian indices contract using $3\otimes\bar{3}$, $2\otimes 2$, and the ϵ-tensor of $\mathrm{SU}(2)$. □

13.4 Problem 4: generation arithmetic

Bouchard-Donagi uses a $\mathbb{Z}_2$ quotient. The chiral index relation gives the number of generations:

$$N_g = \frac{|c_3(\widetilde{V})|}{2|\Gamma|}.$$

Theorem 13.6 (Three-generation cover arithmetic)**.** *If* $|c_3(\widetilde{V})| = 12$ *and* $|\Gamma| = 2$*, the quotient has three generations.*

Proof. Substitution gives

$$N_g = \frac{12}{2 \cdot 2} = 3.$$

□

13.5 Problem 5: Higgs-stratum selection

The Bouchard-Donagi construction has regions with zero, one, or two Higgs doublet conjugate pairs. OPH selects the one-pair stratum.

Theorem 13.7 (One-Higgs zero-mode minimality gate)**.** *Within the published Bouchard–Donagi massless-cohomology strata, the* $n = 1$ *Higgs-pair stratum is the minimal row containing a Higgs pair and no second massless Higgs pair.*

Proof. $n = 0$ lacks a Higgs pair and lacks the Yukawa-complete electroweak branch used by the OPH target. The $n = 2$ stratum has a second massless Higgs pair and therefore fails the declared zero-mode minimality score. The $n = 1$ stratum supplies exactly one massless pair. This proof does not exclude an $n = 2$ low-energy branch: moduli-dependent masses and the full decoupling calculation could lift its additional pair. □

13.6 Problem 6: massless unified-gauge carriers

The connected massless product-group adjoint contains no simple-SU(5) X/Y generator. This Lie-algebra statement does not decide whether massive carriers induce proton decay.

Theorem 13.8 (Product-group adjoint)**.** *The connected adjoint of*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

is

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

It contains no $(3, 2, \pm 5/6)$ *gauge bosons.*

Proof. Dividing by the finite central subgroup $\mathbb{Z}_6$ leaves the Lie algebra unchanged:

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

The adjoint representation decomposes as the adjoints of the three factors:

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

Mixed $(3, 2, \pm 5/6)$ generators occur in the adjoint of simple SU(5) after breaking to the Standard Model subgroup. They are absent from the product Lie algebra above. □

13.7 Problem 7: string emergence

The worldsheet language appears from edge sewing as an effective description.

Theorem 13.9 (Heat-kernel derivation)**.** *For a compact group G, the sewn OPH edge partition*

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}$$

is the heat kernel at the identity:

$$Z_{\text{edge}}(t) = K_t(1).$$

Proof. Peter-Weyl gives the heat kernel expansion

$$K_t(g) = \sum_R d_R \chi_R(g) e^{-tC_2(R)}.$$

At $g = 1$, $\chi_R(1) = d_R$. Substitution gives

$$K_t(1) = \sum_R d_R^2 e^{-tC_2(R)} = Z_{\text{edge}}(t).$$

□

13.8 Problem 8: moduli stabilization as target locking

The OPH contribution is a target criterion. This paper does not emit a completed BD moduli computation.

Theorem 13.10 (Local isolation by full rank)**.** *Let $\mathcal{F} : \mathcal{M} \to \mathbb{R}^K$ be smooth, and suppose*

$$\mathcal{F}(m_\star) = \mathcal{O}_{\text{OPH}}, \qquad \operatorname{rank} D\mathcal{F}(m_\star) = \dim \mathcal{M}_{\text{phys}}$$

after quotienting OPH-stable equivalences. The solution $m_\star$ is locally isolated in the physical moduli directions.

Proof. Work on a local slice transverse to OPH-stable equivalence, of dimension $\dim \mathcal{M}_{\text{phys}}$. The derivative of $\mathcal{F}$ restricted to that slice has full rank. The regular level-set theorem gives a level set of dimension

$$\dim \mathcal{M}_{\text{phys}} - \operatorname{rank} D\mathcal{F} = 0.$$

The target equation has an isolated local solution on the physical slice. □

13.9 Problem 9: operator safety

Gauge invariance allows the standard dangerous operators

$$LH_u, \quad LLe^c, \quad LQd^c, \quad u^c d^c d^c, \quad QQQL, \quad u^c u^c d^c e^c.$$

The operator-safe candidate adds the MSSM $\mathbb{Z}_4^R$ safety layer. Lee, Raby, Ratz, Ross, Schieren, Schmidt-Hoberg, and Vaudrevange prove that, allowing Green-Schwarz anomaly cancellation and requiring a discrete symmetry commuting with $SO(10)$ that forbids the perturbative μ-term, the MSSM has a unique $\mathbb{Z}_4^R$ symmetry. It leaves exact matter parity after nonperturbative breaking and suppresses dimension-five baryon/lepton violation [26].

Use the charge assignment

$$R(Q) = R(u^c) = R(d^c) = R(L) = R(e^c) = 1, \qquad R(H_u) = R(H_d) = 0, \qquad R(W) = 2 \pmod 4.$$

A perturbative superpotential monomial is allowed precisely when its total R-charge is $2 \pmod 4$.

Operator	$\mathbb{Z}_4^R$ **charge**	**Result**
QH_uu^c	$1+0+1=2$	Up-type Yukawa allowed.
QH_dd^c	$1+0+1=2$	Down-type Yukawa allowed.
LH_de^c	$1+0+1=2$	Charged-lepton Yukawa allowed.
LH_uLH_u	$1+0+1+0=2$	Weinberg neutrino operator allowed.
LH_u	$1+0=1$	Bilinear RPV forbidden perturbatively.
LLe^c	$1+1+1=3$	Lepton-number RPV forbidden perturbatively.
LQd^c	$1+1+1=3$	Lepton-number RPV forbidden perturbatively.
$u^cd^cd^c$	$1+1+1=3$	Baryon-number RPV forbidden perturbatively.
$QQQL$	$1+1+1+1=0$	Dimension-five proton decay forbidden perturbatively.
$u^cu^cd^ce^c$	$1+1+1+1=0$	Dimension-five proton decay forbidden perturbatively.
H_uH_d	$0+0=0$	Perturbative μ-term forbidden.

Theorem 13.11 (OPH operator-safety theorem)**.** *At the MSSM charge-algebra level, adding the $\mathbb{Z}_4^R$ layer to $BD_{n=1}^{\mathrm{OPH}}$ permits all Standard Model Yukawa couplings and the Weinberg operator, forbids perturbative dimension-four RPV, forbids perturbative dimension-five proton decay, forbids the perturbative μ-term, and leaves exact matter parity after nonperturbative breaking. The operator-safe candidate is*

$$BD_{n=1,+}^{\mathrm{OPH}} = BD_{n=1}^{\mathrm{OPH}} + \mathbb{Z}_4^R.$$

Proof. The charge table verifies the perturbative selection rule term by term. The allowed Yukawa and Weinberg operators have total R-charge $2 \pmod 4$. The RPV operators, the dimension-five proton-decay operators, and H_uH_d have total charge 1, 3, or $0 \pmod 4$), hence fail the $2 \pmod 4$ superpotential rule. The cited MSSM theorem supplies uniqueness of the $\mathbb{Z}_4^R$ safety layer under the stated anomaly and $SO(10)$-commuting assumptions, plus the matter-parity remnant after nonperturbative breaking. □

14 Acceptance Gates

The package is adversarial by design. The named operator-safe candidate has to pass the same gates as any competitor:

Gate	**Requirement**
Global group	The realized group is $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.
Hypercharge	The charge lattice is the Standard Model lattice.

Color and generations	$N_c = 3$ and $N_g = 3$.
Higgs sector	The low-energy branch has one Higgs pair and decouples to the observed Higgs doublet.
Exotics	No light chiral exotics appear.
Extra visible gauge factors	No extra visible low-scale U(1) appears.
Massless unified-gauge carriers	The connected massless product-group adjoint contains no mixed $(3, 2, \pm 5/6)$ generator.
Operator safety	The MSSM $\mathbb{Z}_4^R$ safety layer forbids perturbative RPV, dimension-five proton decay, and the perturbative μ-term, with exact matter parity after nonperturbative breaking.
Moduli and thresholds	The branch must solve $\mathcal{F}_{BD,n=1,+}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$.

Theorem 14.1 (No massless X/Y connected-adjoint carrier)**.** *On the selected product-group zero-mode branch, the connected massless gauge adjoint is*

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

The adjoint contains no mixed $(3, 2, \pm 5/6)$ *generator.*

This theorem is an algebraic statement about the connected massless OPH product-group adjoint. The Bouchard–Donagi computation is a zero-mode calculation. It does not enumerate massive Wilson-line or Kaluza–Klein X/Y-type modes, calculate their couplings, or bound the proton decay operators obtained by integrating them out. Those questions belong to the heavy-spectrum and threshold certificate.

The local executable package verifies the algebraic gates using exact rational arithmetic:

$$\mathrm{SU}(3)^2\mathrm{U}(1) = 0, \quad \mathrm{SU}(2)^2\mathrm{U}(1) = 0, \quad \mathrm{grav}^2\mathrm{U}(1) = 0, \quad \mathrm{U}(1)^3 = 0.$$

It also verifies the one-Higgs Yukawa hypercharge sums and records the $\mathbb{Z}_2$-cover generation arithmetic

$$|c_3(\widetilde{V})| = 12, \qquad |\Gamma| = 2, \qquad N_g = \frac{|c_3|}{2|\Gamma|} = 3.$$

Remark 14.2 (Operator safety certificate)*.* Gauge invariance leaves the usual MSSM danger operators hypercharge-neutral:

$$LH_u, \quad LLe^c, \quad LQd^c, \quad u^c d^c d^c, \quad QQQL, \quad u^c u^c d^c e^c.$$

Bouchard, Cvetic, and Donagi computed the classical trilinear couplings for this heterotic MSSM and report nonzero up-sector Yukawa couplings, vanishing R-parity-violating terms, and proton stability at that trilinear level [22]. The $\mathbb{Z}_4^R$ layer closes the MSSM charge-algebra safety gate. A BD certificate should realize this safety layer, or an equivalent selection rule, directly in the compactification data.

15 Threshold Proxy Audit and Moduli-Locking Target

The published Bouchard–Donagi calculation fixes a visible zero-mode spectrum. It does not supply the physical vacuum point or solve the nonzero-mode problem. The source inventory contains eleven Kähler moduli, eleven complex-structure moduli, and fifty-one invariant bundle moduli, but no stabilized values or mass matrices [22, 23]. The original strongly coupled completion has a trivial hidden bundle and unbroken hidden E_8; bulk M5-branes cancel the remaining anomaly class. Perturbative hidden-bundle variants exist, so the hidden completion is itself a branch choice [21, 23]. None of these papers supplies a branch-derived string scale, full heavy spectrum, threshold determinant, or low-energy spectrum run.

The OPH low-energy target used here has no supersymmetric partner sector. The interface problem arises because the BD compactification is $N = 1$ supersymmetric. A complete certificate must therefore supply a decoupling map. A conventional route needs supersymmetry breaking, mediation, and soft boundary conditions. A non-supersymmetric route would instead require a new, independently consistent UV construction or deformation. To certify this BD candidate, it must be proved an OPH-equivalent continuation that preserves the cited BD visible-branch data, as well as establishing worldsheet or modular consistency, anomaly cancellation, a stable vacuum, its spectrum and couplings, and the resulting thresholds. An unrelated construction is a different candidate, and a projection label or hash is not sufficient.

The quantities below are target-side one-loop proxies. They do not use a BD heavy spectrum, stabilized moduli point, decoupling map, hidden-sector dynamics, mediation data, soft terms, or spectrum-generator output. Proxy reproduction is therefore not threshold compatibility, and $BD^{\mathrm{OPH}}_{n=1,+}$ remains a named candidate rather than a passing OPH-correct witness. The machine-readable bundle under `code/particles/runs/uv/oph_bd_threshold_spectrum/` records repository-relative dependency paths, SHA-256 hashes, JSON selectors, an 80-decimal-digit arithmetic policy, scheme status, and the fail-closed gate result.

From frozen target-side coordinates, including candidate-only Higgs/top inputs, the proxy computes

Quantity	**Proxy coordinate from declared inputs**
Tree-level top coordinate	$y_t^{\mathrm{tree}} = 0.987745211164$
Tree-level Higgs quartic coordinate	$\lambda_H^{\mathrm{tree}} = 0.128706603202$
MSSM tree-level quartic ceiling	$\lambda_{\mathrm{MSSM,max}} = 0.068973725409$
Large-tan β algebraic quartic gap	$\Delta\lambda_{\mathrm{proxy}} = 0.059732877792$
Tree-level Higgs mass proxy ceiling	$m_{h,\mathrm{tree}}^{\mathrm{max}} = 91.6524602856\,\mathrm{GeV}$

Equivalently, the tree-level coordinates are

$$y_t^{\mathrm{tree}} = 0.987745211164, \qquad \lambda_H^{\mathrm{tree}} = 0.128706603202,$$

with the MSSM tree-level quartic ceiling

$$\lambda_{\mathrm{MSSM,max}} = 0.068973725409,$$

and the corresponding algebraic gap is

$$\Delta\lambda_{\mathrm{proxy}} = 0.059732877792.$$

The displayed y_t^{tree} is likewise the algebraic pole-coordinate conversion $\sqrt{2}m_t/v$, not a running Yukawa in a common scheme. The quartic subtraction mixes declared surface or pole coordinates with a tree-level MSSM expression and does not define a running quartic in a common renormalization scheme.

15.1 One-loop stop proxy for the conventional breaking route

Using

$$\Delta m_h^2 \simeq \frac{3m_t^4}{2\pi^2 v^2}\left[\log\frac{M_S^2}{m_t^2} + \frac{X_t^2}{M_S^2}\left(1 - \frac{X_t^2}{12M_S^2}\right)\right],$$

and the tree-level electroweak ceiling above, the frozen proxy gives:

$\tan\beta$	X_t/M_S	M_S proxy target
2	0	3046.266 GeV
2	$\sqrt{6}$	679.714 GeV
5	0	1191.830 GeV
5	$\sqrt{6}$	265.933 GeV
10	0	968.658 GeV
10	$\sqrt{6}$	216.137 GeV
50	0	901.608 GeV
50	$\sqrt{6}$	201.176 GeV

These rows invert a leading one-loop formula. They do not determine $\tan\beta$, M_S, or X_t from the BD branch, and some rows have too little scale separation for a controlled leading-log interpretation. If the conventional breaking route is chosen, a full soft-term and spectrum calculation must test them in one running scheme. A non-supersymmetric alternative must derive its own masses and decoupling thresholds within the independently consistent construction.

15.2 Gauge-unification proxy

A one-loop SM/MSSM running proxy with $M_{\mathrm{SUSY}} = 1\,\mathrm{TeV}$ and $\alpha_1 = (5/3)\alpha_Y$ gives

$$\log_{10}(M_U/\mathrm{GeV}) = 16.0815812332,$$

$$M_U = 1.20665 \times 10^{16}\,\mathrm{GeV}, \qquad \alpha_U^{-1} = 26.0176807530,$$

$$\alpha_3^{\mathrm{pred}}(m_Z) = 0.111511310319.$$

Against the PDG 2025 comparison value $\alpha_s(m_Z) = 0.1180 \pm 0.0009$ [25], the residual inverse-coupling combination is

$$\mathcal{C}_3^{\mathrm{proxy}} = -0.493123930167,$$

with the reference-only interval

$$-0.557271269478 \le \mathcal{C}_3^{\mathrm{proxy}} \le -0.427990551481.$$

Here the declared matching convention is

$$\mathcal{C}_3^{\mathrm{proxy}} = \alpha_s(m_Z)^{-1} - \left[\alpha_3^{\mathrm{pred}}(m_Z)\right]^{-1}$$

after the one-loop SM/MSSM step running. This is not a computed BD string/GUT threshold. Scanning the arbitrary common M_{SUSY} step from 0.5 to 10 TeV moves the central residual from about -0.3434 to -0.9905. A heavy spectrum and a fixed matching prescription are needed before a physical threshold corridor exists.

15.3 Encoded candidate sieve

The threshold-proxy audit scores representative branches against the encoded structural gates. The threshold and moduli gates are certificate gates, not part of this finite structural score:

Candidate	Score	Verdict
$BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=0}$	7/9	Rejected: no Higgs pair.
$BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=1}$	8/9	Geometric witness; safety layer absent.
$BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=1,+}$	9/9	Selected inside the nine structural gates only; not an OPH-correct witness.
$BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=2}$	7/9	Fails the zero-mode minimality score.
BHOP $\mathrm{SU}(4), \mathbb{Z}_3 \times \mathbb{Z}_3$	4/9	Backup witness.
Generic $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic	0/9	Rejected as minimal class.

Inside this encoded structural audit, $BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=1}$ is the selected geometric zero-mode witness and $BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=1,+}$ is the selected operator-safe candidate. The score excludes the threshold and moduli gates and therefore cannot establish OPH-correctness.

Theorem 15.1 (Encoded structural-only audit uniqueness)**.** *Let $\mathcal{C}_{\mathrm{audit}}$ be the six-branch candidate family in the table above, scored against the nine encoded structural gates. The unique full-score row in $\mathcal{C}_{\mathrm{audit}}$ is $BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=1,+}$. No threshold, moduli, or low-energy passage follows from this finite score.*

Proof. The table assigns scores $7/9, 8/9, 9/9, 7/9, 4/9, 0/9$ to the six listed candidates. The only score equal to the structural gate count is the $BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=1,+}$ row. The full-score structural set inside $\mathcal{C}_{\mathrm{audit}}$ is the singleton

$$\{BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=1,+}\}.$$

□

These proxy numbers define a target-side screen for a full cohomology, Yukawa, heavy-spectrum, threshold, and decoupling computation. The conventional MSSM route requires supersymmetry breaking and soft boundary conditions. A non-supersymmetric route can certify this BD candidate only if it is an independently consistent, OPH-equivalent deformation or continuation that preserves the cited BD visible-branch data; an unrelated construction is a different candidate. Cohomology algorithms such as cohomCalg-style line-bundle cohomology are designed for massless-mode calculations in compactifications [31]. Spectrum tools such as SOFTSUSY solve MSSM renormalization-group equations with supplied conventional breaking constraints [32]; citing such a tool is not evidence that a BD spectrum run occurred.

Theorem 15.2 (OPH moduli-locking criterion)**.** *Let $\mathcal{M}_{BD,n=1,+}$ be the moduli, threshold, safety-layer, and decoupling parameter space of the selected one-Higgs operator-safe stratum, and let*

$$\mathcal{F}_{BD,n=1,+} : \mathcal{M}_{BD,n=1,+} \to \mathbb{R}^K$$

send a point m to its gauge, Yukawa, Higgs, threshold, and mass vector. If

$$\mathcal{F}_{BD,n=1,+}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$$

and

$$\operatorname{rank} D\mathcal{F}_{BD,n=1,+}(m_\star) = \dim \mathcal{M}_{\mathrm{phys}}$$

on a slice transverse to OPH-stable equivalence, then $m_\star$ is locally isolated as a physical solution of the target equation.

16 Conditional BD Witness Certificate Criterion

The strongest mathematically clean statement has certificate form. It separates what the paper proves by algebra from what the BD computation would have to certify. The required certificate is not emitted here, and no global uniqueness statement follows without a complete comparison catalogue.

Theorem 16.1 (Conditional OPH BD witness implication)**.** *Assume the following data.*

1. *The OPH recovered-core target is*

$$\mathfrak{S}_{\mathrm{OPH}} = \begin{pmatrix} (\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6, \quad N_c=3, \quad N_g=3, \quad Y_{\mathrm{SM}}, \\ n_H=1, \quad \textit{no light chiral exotics}, \quad \mathbb{Z}_4^R \textit{ safety} \end{pmatrix}.$$

2. *The effective string continuation is read through the heterotic edge-sector branch and satisfies the finite-carrier critical-edge certificate of Section 9.*

3. *The BD cohomology certificate realizes the massless one-Higgs, three-generation, no-visible-exotics stratum with Wilson-line centralizer $S(\mathrm{U}(3)\times\mathrm{U}(2))$.*

4. *The same branch realizes $\mathbb{Z}_4^R$, or an OPH-equivalent safety rule, on the visible operator algebra.*

5. *A threshold/spectrum receipt supplies the nonzero-mode and hidden spectra, the dilaton and other stabilized moduli, the bulk five-brane sector or a proof of its absence, normalized Yukawa boundary data, physical scales, and a map that decouples the supersymmetric compactification into the non-supersymmetric low-energy branch. For a conventional breaking route this includes mediation and soft boundary conditions. A non-supersymmetric alternative must instead supply an independently consistent, OPH-equivalent deformation or continuation of the BD branch, prove that it preserves the cited visible cohomology and safety data, and compute its threshold map. An unrelated construction does not certify the BD candidate.*

6. *There is $m_\star \in \mathcal{M}_{BD,n=1,+}$, on the complete source surface described in Section 18, with*

$$\mathcal{F}_{BD,n=1,+}(m_\star) = \mathcal{O}_{\mathrm{OPH}}, \qquad \operatorname{rank} D\mathcal{F}_{BD,n=1,+}(m_\star) = \dim \mathcal{M}_{\mathrm{phys}}.$$

Then the following OPH-stable equivalence class is a passing named critical-string witness:

$$\boxed{\left[BD_{n=1}^{\mathrm{SU}(5),\mathbb{Z}_2} + \mathbb{Z}_4^R\right]_{\mathrm{OPH}} = BD_{n=1,+}^{\mathrm{OPH}}.}$$

Proof. The recovered-core target fixes the observer-visible group, hypercharge lattice, generation count, Higgs count, exotic-matter gate, and operator-safety gate. The heterotic edge-sector branch and the finite-carrier critical-edge certificate place the named witness on the critical edge route. The BD certificate supplies a nonempty $\mathbb{Z}_2$-quotient SU(5)-bundle witness with the required three-generation, no-exotics, one-Higgs massless cohomology. The centralizer and quotient theorems identify its Wilson-line centralizer with the OPH global Standard Model group. The $\mathbb{Z}_4^R$ safety layer closes the visible operator algebra gate. The threshold/spectrum and decoupling hypothesis supplies the physical map from that supersymmetric zero-mode branch to the OPH target. The moduli-locking rank hypothesis isolates $m_\star$ modulo OPH-stable equivalence. Therefore the named class

$$\left[BD_{n=1}^{\mathrm{SU}(5),\mathbb{Z}_2} + \mathbb{Z}_4^R\right]_{\mathrm{OPH}}$$

satisfies the OPH-correct witness gates. □

Remark 16.2 (Certificate boundary). The theorem gives the witness proof shape. Its algebraic steps close in this paper. The critical-edge receipt table, cohomology table, $\mathbb{Z}_4^R$-realization table, Yukawa/superpotential table, heavy-spectrum and decoupling table, threshold table, and Jacobian-rank table would form the external witness package. That package is not emitted here. A comparison table over an exhaustive candidate class is an additional requirement.

Theorem 16.3 (Certificate closure criterion)**.** *The class-restricted equality*

$$\{\mathcal{T} \in \mathcal{C} : \mathcal{T} \text{ is OPH-correct}\}/\sim_{\mathrm{OPH}} = \{BD_{n=1,+}^{\mathrm{OPH}}\}$$

is fully certified for an explicitly enumerated candidate class $\mathcal{C}$ once the seven certificates in Section 18 verify: critical-edge closure, cohomology, $\mathbb{Z}_4^R$ realization, superpotential safety, threshold matching, moduli locking with full transverse rank, and comparative uniqueness. If any of the first six certificates fails, the named witness retracts. If the seventh certificate fails, uniqueness retracts and the surviving OPH-admissible set requires a wider comparison audit. Without an exhaustive $\mathcal{C}$, the criterion does not assert a global critical-string singleton.

Proof. The criterion is a conjunction of seven independent certificate gates. Gates 1–6 are existence and safety gates for the named witness. Gate 7 is a comparison gate over the declared candidate class. If all seven hold for an exhaustive $\mathcal{C}$, the named witness exists and is the unique minimal OPH-admissible survivor inside $\mathcal{C}/\sim_{\mathrm{OPH}}$. If an existence or safety gate fails, the named witness misses the definition of OPH-correctness. If the comparison gate alone fails, the named witness may remain OPH-correct, with singleton claim retracted. This proves the criterion. □

17 Empirical Test Surface

The candidate is useful because its structural claims and assigned proxy targets can fail. The table keeps those two classes separate. Published zero-mode results test the visible massless branch. Numerical rows define screens for the required heavy-spectrum and low-energy calculation, which is not emitted here; they are not realized BD predictions.

Structural claim or proxy screen	**Numerical or structural target**	**Failure meaning**

Global Standard Model group	$(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.	Retracts the named visible branch.
Color and generations	$N_c = 3$, $N_g = 3$, with $\lvert c_3(\widetilde{V})\rvert = 12$, $\lvert\Gamma\rvert = 2$.	Retracts the BD witness.
Higgs sector	One massless Higgs pair in the published cohomology; its decoupling to the observed light Higgs requires a separate map.	Extra massless pairs retract the selected zero-mode stratum; a failed decoupling map blocks low-energy passage.
Visible exotics	No massless visible chiral exotics in the published cohomology.	Retracts the minimal zero-mode branch.
Visible gauge factors	No extra visible low-scale U(1).	Severe pressure on the normal-form sieve.
Massless unified-gauge carriers	No mixed $(3, 2, \pm 5/6)$ generator in the connected massless product-group adjoint.	A massless mixed generator conflicts with the product-group branch; massive BD modes require the heavy-spectrum audit.
Operator safety	$\mathbb{Z}_4^R$ permits Yukawas and the Weinberg operator, forbids perturbative RPV, dimension-five proton decay, and the perturbative μ-term, and leaves matter parity after nonperturbative breaking [26].	Safety certificate fails if the BD branch lacks this layer or an equivalent rule.
Classical RPV trilinears	BCD report vanishing R-parity-violating trilinears at the classical level in this heterotic MSSM [22]. Worldsheet-instanton corrections remain outside that result.	Classical superpotential audit failure.
Tree-level top coordinate	$y_t^{\mathrm{tree}} = 0.987745211164$.	A completed forward spectrum misses the assigned proxy screen; no failure is inferred from the proxy alone.
Tree-level Higgs coordinate	$\lambda_H^{\mathrm{tree}} = 0.128706603202$.	A completed same-scheme calculation misses the assigned proxy screen.
Algebraic quartic gap	$\Delta\lambda_{\mathrm{proxy}} = 0.059732877792$.	A conventional breaking calculation fails this proxy screen; the row is not a heavy-spectrum certificate.

Stop proxy	The table gives M_S values for representative $(\tan\beta, X_t/M_S)$ choices.	Conventional-route soft terms miss the proxy screen. A BD-equivalent non-supersymmetric deformation has its own mass and threshold calculation.
One-loop gauge residual	At $M_{\rm SUSY} = 1\,\text{TeV}$, $\mathcal{C}_3^{\rm proxy} = -0.493123930167$, with the reference-only interval shown in Section 15.	A supplied heavy-threshold combination misses the proxy screen; the residual itself is not a BD string threshold.

The published branch supplies a narrow massless visible package. This paper assigns numerical screens, but the required calculation is not emitted here; no threshold corridor is established for the BD candidate. The open map is

$$\mathcal{F}_{BD,n=1,+}(m_\star) = \mathcal{O}_{\rm OPH}.$$

18 Claim Gates

The computation package reduces the full string-model task to explicit inputs. The gate table is the working contract for an independent heterotic reproduction:

Gate	**Needed input**	**State of the paper claim**
Central-charge and supercurrent certificate	Response-map rank $\operatorname{rank}\mathcal{R}_{\rm port} = 12$, positive $A_5 \times C_2$ covariance sectors, edge transfer OPEs, local graded half-repair $Q_N^2 \to H_R$, and torus/spin-structure receipt.	Non-implication theorem and finite critical-edge certificate theorem stated; carrier receipts not emitted here.
Cohomology certificate	Explicit X, V, and equivariant $\mathbb{Z}_2$ action in cohomCalg, Sage, or Macaulay2 form.	Structural witness cited; raw computation not emitted here.
Safety-layer realization certificate	$\mathbb{Z}_4^R$, or an equivalent compactification selection rule, realized on the BD one-Higgs branch.	MSSM charge algebra closes; BD realization certificate not emitted here.
Operator and superpotential certificate	Sheaf cup products, harmonic representatives, instanton data, and sector selection rules.	BCD trilinear result supports RPV vanishing; numeric textures not emitted here.

Threshold and spectrum certificate	Nonzero-mode and hidden spectra; stabilized Kähler, complex-structure, bundle, dilaton, and any bulk five-brane data; normalized Yukawa boundary data; string, compactification, and matching scales; and a low-energy decoupling map. A conventional breaking route also requires mediation and soft boundary conditions. A non-supersymmetric alternative requires an independently consistent UV construction or deformation and its own mass and threshold calculation.	**Not emitted here:** proxy coordinates reproduce, but compatibility is not evaluated. No heavy-threshold or low-energy spectrum certificate is present.
Moduli-locking certificate	Full map $D\mathcal{F}_{BD,n=1}$ on the physical moduli slice.	Isolation criterion proved; rank computation not emitted here.
Comparative uniqueness certificate	Expanded heterotic edge-sector audit, plus any brane, orientifold, or duality-rewritten presentations claiming OPH equivalence.	Encoded audit selects $BD^{\mathrm{OPH}}_{n=1,+}$ inside its declared rows; broader catalogue audit not emitted here.

Without passing threshold/spectrum and moduli-locking receipts, $BD^{\mathrm{OPH}}_{n=1,+}$ is a structural and operator-safe candidate. It is not a passing OPH-correct witness.

19 Falsifier Matrix

Failure mode	**Consequence**
Wrong global Standard Model group	Retracts the OPH visible landing branch.
Wrong hypercharge lattice	Retracts the named visible branch.
Fourth chiral generation	Retracts the realized branch.
Light chiral exotics	Retracts the BD one-Higgs witness.
Irreducible second light Higgs pair	Retracts the one-Higgs low-energy projection.
Extra visible low-scale U(1)	Severe pressure on the OPH normal-form sieve.
Massless mixed X/Y generator	Conflicts with the connected product-group branch.
Massive BD X/Y-type modes or induced operators	Evaluated by the heavy-spectrum and coupling certificate; the zero-mode adjoint theorem does not decide them.
BD cohomology reproduction fails	Retracts $BD^{\mathrm{OPH}}_{n=1}$ as named geometric witness.
$\mathbb{Z}_4^R$ or equivalent safety layer absent	Retracts $BD^{\mathrm{OPH}}_{n=1,+}$ as operator-safe candidate.

Dangerous operators survive without suppression	Severe pressure on the effective theory.
No moduli point matches $\mathcal{O}_{\mathrm{OPH}}$	Retracts the operator-safe candidate.
Central-charge or supercurrent gate fails	Retracts the heterotic critical-worldsheet identification of the sewn edge branch.
Critical-string lift fails	Weakens the string continuation; the OPH recovered core is a separate claim tier.

20 Proof Stack for De Sitter and String Theory

The most persuasive route is compact:

1. prove the finite observer theorem as stable checkpoint continuation;
2. prove the clock-projection correlator theorem in finite dimension;
3. prove T-holonomy as a $\mathbb{Z}_2^T$ cycle obstruction and $w_1(L_T)$ class;
4. prove record-branch time-reversal breaking on the observer quotient;
5. derive $S_{\mathrm{dS}} = A/(4\ell_P^2) = N_{\mathrm{scr}}$ as edge-center record capacity;
6. derive $Z_{\mathrm{edge}}(t) = K_t(1)$ and the large-N_{edge} worldsheet branch;
7. prove the heterotic edge-polarization theorem for the sewn-edge scaling limit, including OPEs, fermionic grading, spin structures, and no residual coset;
8. prove the OPH vacuum sieve, global group lock, $\mathbb{Z}_4^R$ safety table, connected-massless- adjoint theorem, and moduli-locking criterion;
9. reproduce the Bouchard-Donagi $n = 1$ cohomology, $\mathbb{Z}_4^R$ realization, and operator catalogue;
10. freeze the nonzero-mode and hidden spectra, all physical moduli and scales, either a conventional SUSY-breaking decoupling route or an independently consistent OPH-equivalent non-supersymmetric deformation with its own mass and threshold calculation, the low-energy forward calculation, threshold matching, repository-relative source paths and hashes, and numerical precision; then evaluate the target map and its physical-moduli Jacobian.

The first two items target the de Sitter observer and time-reversal program. The last four items target the string community's demand for a reproducible critical-worldsheet and vacuum selector.

21 Problem-to-Gate Map for De Sitter and String Theory

The paper reduces the program to a finite observer-visible normal-form problem for critical strings. In that form, many familiar unresolved questions become gates with explicit pass/fail certificates.

Problem	**OPH resolution**	**Certificate gate**
What is the physical observer in de Sitter?	A stable record-bearing patch subfederation with checkpoint continuation.	Record/collar model.

Why does a clock branch appear?	Clock time is a record-sector projection; imaginary semiclassical correlators appear after branch conditioning.	Observer-sector examples.
What is the clock-flip holonomy?	A $\mathbb{Z}_2^T$ cycle obstruction on the overlap nerve, $w_1(L_T)$.	Geometric examples.
Where is de Sitter entropy?	Edge-center record capacity, $S_{\mathrm{dS}} = A/(4\ell_P^2) = N_{\mathrm{scr}}$.	Capacity normalization.
Why do strings appear?	Sewn OPH edge partitions are heat-kernel sums; the controlled large-edge continuation is a worldsheet expansion.	Large-edge control.
Which string theory passes the OPH selector?	No candidate is promoted by this paper. $BD^{\mathrm{OPH}}_{n=1,+}$ is the named conditional candidate. Passage requires every stated gate; global uniqueness also requires an exhaustive comparison audit.	Critical-edge, cohomology, safety, threshold/spectrum, decoupling, and moduli certificates.
Why anthropic landscape selection is absent	OPH provides a target normal form $\Pi^{\mathrm{OPH}}_{\mathrm{IR}}(\mathcal{T}) = \mathfrak{S}_{\mathrm{OPH}}$, with no statistical sampling weight.	Exhaustive audits.
Why the exact SM global group?	Wilson-line centralizer and OPH quotient both give $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.	Algebraic proof in paper.
Why no massless simple-GUT proton-decay carrier?	The connected product-group adjoint has no $(3, 2, \pm 5/6)$ gauge bosons. Massive BD modes and induced operators remain in the heavy-spectrum gate.	Algebraic massless-adjoint proof plus heavy-mode certificate.
How is the μ-problem controlled?	The $\mathbb{Z}_4^R$ safety layer forbids perturbative $H_u H_d$ and allows nonperturbative generation.	Compactification realization.
How are moduli stabilization claims tested?	The candidate must solve the target-locking equation $\mathcal{F}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$ with full transverse rank. No solution point is emitted.	Numerical/rank certificate.
What is the string graviton?	The closed-string vertex image of the OPH transverse-traceless metric perturbation.	Normalization match.
What fixes the critical dimension?	The critical-edge certificate must identify the sewn-edge carrier with the heterotic edge VOA; then heterotic criticality gives $D = 10$ and $c_{\mathrm{internal}} = 16$.	Edge transfer, OPE, grading, spin-structure, and no-coset receipts.

Theorem 21.1 (De Sitter/string compression theorem)**.** *On the OPH-correct branch, the de Sitter observer problem, the time-reversal clock-branch problem, the entropy-location problem, and the critical-string vacuum-selection problem reduce to one finite statement: find the observer-visible normal form of the OPH patch federation and then lift its sewn edge sector to a critical worldsheet presentation.*

Proof. The observer and clock problems are properties of record-bearing subfederations and their $\mathbb{Z}_2^T$ cycle data. The entropy problem is the edge-center capacity of the same finite static patch.

The string problem is the critical completion of the sewn edge-sector partition. The OPH normal form fixes the visible quotient data used by all four questions. The apparently separate problems are projections of the same finite patch-federation normal-form problem. □

22 Conclusion

OPH gives a finite observer/record/overlap implementation layer for de Sitter holography. The same edge system admits a sewn-worldsheet language. Critical string theory enters as the effective completion of that language only after the critical-edge certificate closes. Conditional on that gate, the OPH visible-normal-form target tests the Bouchard–Donagi one-Higgs heterotic Standard Model as a geometric zero-mode candidate. Adding the MSSM $\mathbb{Z}_4^R$ safety layer gives the named operator-safe candidate

$$BD_{n=1,+}^{\mathrm{OPH}} = BD_{n=1}^{\mathrm{OPH}} + \mathbb{Z}_4^R.$$

The threshold/spectrum certificate is not emitted here because the published BD data do not contain a stabilized heavy spectrum or a map from the supersymmetric compactification to the non-supersymmetric OPH low-energy branch. The critical-edge, cohomology, safety-layer, threshold, decoupling, and moduli gates therefore leave $BD_{n=1,+}^{\mathrm{OPH}}$ at candidate status.

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